A LOCAL APPROXIMATION CHARACTERIZATION OF OKA MANIFOLDS

SICHENG AN, YUN-HENG DU, BIN GUO, PENG-CHAO WANG, AND SONG-YAN XIE

Abstract. We introduce the local approximation property (LAP) and prove that it characterizes Oka manifolds. We prove that LAP yields convex approximation and, in an n-dimensional target, propagates across sets of zero $(2n-1)$-dimensional Hausdorff measure. Consequently, strong dominability characterizes Oka manifolds, convex approximation in the target dimension suffices, the Oka property is local in the Euclidean topology, and it descends under surjective holomorphic maps. Local families of entire curves give geometric applications: smooth projective rationally connected manifolds are Oka, and deleting an algebraic subset of codimension at least two from any such manifold preserves the Oka property. Structure theorems further show that compact Kähler manifolds with nonnegative holomorphic sectional curvature or nef tangent bundle are Oka. Finally, the complement of a smooth degree-d hypersurface in $\mathbb{P}^n$ is Oka exactly when $d \leq n+1$.

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Date: October 14, 2026.
2020 *Mathematics Subject Classification.* 32Q56, 32E30, 14M22, 14J45.
Key words and phrases. Oka manifold, strong dominability, convex approximation, rational connectedness, logarithmic rational curve.

1. Introduction

The Oka principle concerns the existence of holomorphic solutions to problems on Stein spaces when there is no topological obstruction, provided the target is flexible enough. It is not a single theorem but a program: one looks for geometric conditions on a complex manifold Y under which continuous maps from Stein spaces into Y can be deformed to holomorphic maps, with approximation, interpolation, and parametric control. For broad accounts, see the survey of Forstnerič and Lárusson [28] and Forstnerič's recent exposition [26]. The subject began with Oka's solution of the second Cousin problem [50] and Grauert's theorem for holomorphic fiber bundles with homogeneous fibers over Stein spaces [39]. Gromov placed it in the framework of the h-principle and introduced ellipticity through dominating holomorphic sprays [40], and Forstnerič introduced subellipticity and proved the Oka principle for subelliptic manifolds [20]. Forstnerič then established approximation, interpolation, and finite-order jet interpolation [22], identified convex approximation with the Oka property with approximation [23], and proved that convex approximation implies the full parametric Oka property [24]. This is the notion of an Oka manifold used below; see also Forstnerič's monograph [25].

Here we show that this global holomorphic flexibility is detected by a local approximation operation: centered holomorphic families on expanding discs characterize Oka manifolds. The characterization also shows that the Oka property propagates across proper analytic subsets and descends under surjective holomorphic maps from Oka manifolds, leading to applications to projective manifolds and complements.

Definition 1.1. Let Y be a complex manifold with a distance d_Y inducing its topology. The manifold Y is an *Oka manifold* if it has the *convex approximation property* CAP [24, Definition 1.2]: for every $m \geq 1$, compact convex set $K \subset \mathbb{C}^m$, holomorphic map $f : U \to Y$ on an open neighborhood U of K, and $\epsilon > 0$, there is a holomorphic map $F : \mathbb{C}^m \to Y$ such that

$$\sup_{z \in K} d_Y(F(z), f(z)) < \epsilon.$$

By Forstnerič's theorem [24, Theorem 1.1], this is equivalent to the parametric Oka property, with approximation and interpolation, for maps from reduced Stein spaces.

For $r > 0$, write $\Delta_r = \{t \in \mathbb{C} : |t| < r\}$.

Definition 1.2. Let Y be a complex manifold of dimension $n \geq 1$. One says that Y satisfies the *local approximation property* (LAP) at $x \in Y$ if there are an open neighborhood $V \ni x$, a number $\tau > 0$, radii $R_\nu \to \infty$, and holomorphic maps

$$h_{j,\nu} : V \times \Delta_{R_\nu} \longrightarrow Y, \qquad h_{j,\nu}(y, 0) = y, \qquad 1 \leq j \leq n,$$

converging locally uniformly on $V \times \Delta_\tau$ to maps h_j, such that

$$\partial_t h_1(x, 0), \ldots, \partial_t h_n(x, 0)$$

form a basis of $T_x Y$. One says that Y satisfies LAP if it satisfies LAP at every point. In dimension zero, LAP holds at every point.

Write $G_Y \subset Y$ for the points at which Y satisfies LAP. By continuity of the initial velocities, the locus G_Y is open: after shrinking V, the same families satisfy the definition at every point of V.

All the data in this definition are common to the finitely many families, after restriction and omission of a finite initial segment if necessary. In particular, V is independent of ν, and convergence is imposed on the fixed product $V \times \Delta_\tau$ without requiring an entire limiting map. A finite spanning collection can always be reduced to a basis.

Theorem 1.3. *A complex manifold is Oka if and only if it satisfies* LAP.

The implication LAP $\Rightarrow$ CAP proceeds from local planar extension to polydisc approximation and then to arbitrary compact convex sets by induction across real faces. The local extension retains any prescribed passive parameter polydisc, so that earlier source and spray variables can serve as holomorphic parameters at subsequent stages. Alarcón and Forstnerič [2, Definition 7.11] introduced the local spray approximation property (LSAP) in their study of Oka-1, the Oka property with approximation and jet interpolation from open Riemann surfaces.

An n-dimensional manifold Y is *dominable at* y if there is a holomorphic map $H : \mathbb{C}^n \to Y$ with $H(0) = y$ and dH_0 invertible, and is *strongly dominable* if this holds at every point. Dominability at y gives LAP at y by a local inverse of H (Proposition 2.2), while every Oka manifold is strongly dominable by Forstnerič's result [25]. Hence Theorem 1.3 settles the strong-dominability question raised by Forstnerič and Lárusson [30] and subsequently emphasized by Forstnerič [26].

Theorem 1.4. *A complex manifold is Oka if and only if it is strongly dominable.*

The next three corollaries concern source dimension, stratifications, and open covers. For $d \geq 1$, write CAP_d for convex approximation on compact convex subsets of $\mathbb{C}^d$.

Corollary 1.5. *Let X be an n-dimensional complex manifold, where $n \geq 1$. Then X is Oka if and only if it satisfies* CAP_n.

Forstnerič and Lárusson introduced a *finite Oka stratification*: a finite decreasing filtration by closed complex subvarieties, ending in the empty set, whose successive differences are manifolds with Oka connected components [29, Definition 1]. They proved that such a stratification implies strong dominability [29, Theorem 2].

Corollary 1.6. *Every complex manifold with a finite Oka stratification is Oka.*

Corollary 1.7. *If every point of a complex manifold has an open Oka neighborhood, then the manifold is Oka.*

Corollary 1.5 resolves Forstnerič's stabilization question at the proposed threshold $\dim X$ [23, Problem 6.2]. Corollary 1.6 shows that a finite Oka stratification gives an Oka manifold, a question raised by Forstnerič and Lárusson [29, p. 3716]. Kusakabe proved localization for Zariski-open covers and asked whether Euclidean-open covers suffice [44]; Corollary 1.7 settles his question. The proofs are given in Subsection 6.5.

Kusakabe's Ell_1 characterization requires every holomorphic map $f : S \to Y$ from a Stein manifold to admit a dominating spray $F : S \times \mathbb{C}^N \to Y$ for some N, with $F(\cdot, 0) = f$; sprays over maps near compact convex sets also suffice [44]. Theorem 1.4 obtains the Oka property from the pointwise case, with S a singleton. Kusakabe also characterized Oka manifolds by dense entire curves in spaces of holomorphic maps and by universal maps [43].

For $s \geq 0$, let $\mathcal{H}^s$ denote s-dimensional Hausdorff outer measure with respect to a smooth Riemannian metric. Its null sets are independent of this choice, since smooth metrics are locally bi-Lipschitz equivalent.

Theorem 1.8. *Let Y be a complex manifold of dimension $n \geq 1$ and $E \subset Y$ a set satisfying $\mathcal{H}^{2n-1}(E) = 0$. If Y satisfies* LAP *at every point of $Y \setminus E$, then it satisfies* LAP *everywhere and is Oka, without assuming that E is closed.*

Dominability outside such a set therefore implies the full Oka property. This strengthens the Oka-1 conclusion of Alarcón and Forstnerič [2], which assumes the exceptional set to be closed. Forstnerič and Lárusson developed further properties of Oka-1 manifolds and of LSAP [30]. For analytic exceptional sets, Theorem 1.8 has the following direct consequence:

Theorem 1.9. *Let Y be a complex manifold. Suppose that every point of Y has a connected open neighborhood U and a proper closed complex analytic subset $A_U \subset U$ such that Y satisfies* LAP *at every point of $U \setminus A_U$. Then Y satisfies* LAP *everywhere and is Oka. In particular, if X is a connected complex manifold and $A \subset X$ is a proper closed complex analytic subset such that $X \setminus A$ is Oka, then X is Oka.*

Theorem 1.9 settles Forstnerič's union problem [37, Problem 3.11] and the first part of Question D of Forstnerič and Lárusson [28]: one nonempty Zariski-open Oka subset suffices.

The preceding criterion for exceptional sets of zero $(2n-1)$-dimensional Hausdorff measure also gives descent along arbitrary surjective holomorphic maps from Oka manifolds (Theorem 7.5). Together with Forstnerič's surjectivity theorem [32], this characterizes positive-dimensional connected Oka manifolds by surjective entire maps (Corollary 7.7) and settles Forstnerič's Problem 1.4(a).

Geometric applications. A smooth complex projective manifold X is *rationally connected* if a general pair of its points lies on the image of a map $\mathbb{P}^1 \to X$, and is *Fano* if K_X^{-1} is ample. Deformations of very free rational curves give:

Theorem 1.10. *Every smooth connected complex projective rationally connected manifold is Oka. In particular, every smooth complex Fano manifold is Oka.*

Theorem 1.10 settles the rationally connected case of Forstnerič's Problem 11.3 [37]. In particular, every smooth hypersurface of degree $d \le n$ in $\mathbb{P}^n$ is Oka, which gives the affirmative answer to the hypersurface part of Problem E of Forstnerič and Lárusson [28] throughout the Fano range.

Campana and Winkelmann constructed dense entire curves in rationally connected projective manifolds whose images contain prescribed countable subsets [8]. Alarcón and Forstnerič formulated the Oka-1 conjecture for these manifolds [2], and Du, Guo, and Xie proved it [17]. Related analytic constructions of rational curves were developed by Du, Guo, and Xie [16]. Among earlier full Oka results are the smooth quartic hypersurfaces of dimension at least three [38] and all smooth Fano threefolds [4]. Section 8 proves the general rationally connected case directly by constructing families of entire curves with independent tangent directions.

Corollaries 8.6 and 8.7 show that compact Kähler manifolds with nonnegative holomorphic sectional curvature or nef tangent bundle are Oka. Forstnerič and Kusakabe proved that, for $n \ge 2$ and $k \ge 1$, the unit disc bundle defined by a semipositive Hermitian metric on $\mathcal{O}_{\mathbb{P}^n}(k)$ is Oka [27].

For complements, Kusakabe's theorem covers compact polynomially convex sets in $\mathbb{C}^n$, $n > 1$ [45]; Forstnerič and Wold treated broader classes of unbounded convex sets [34], and further examples come from the periodic-transport constructions of Du, Guo, Wang, and Xie [15]. In the projective setting, Theorem 9.1 gives a criterion for $X \setminus D$ using a rational curve $f : \mathbb{P}^1 \to X$ with $f^{-1}(D) \subseteq \{\infty\}$ and ample $f^*T_X(-\log D)$, for a reduced simple normal crossings divisor D in a connected projective manifold X. Here $T_X(-\log D)$ is the bundle of vector fields tangent to every component of D. Chen and Zhu's theory of $\mathbb{A}^1$-curves on log smooth varieties [14], combined with Corollary 9.4, gives applications to ample rationally connected divisors and complete intersections. The same deformation method also shows that deleting an algebraic subset of codimension at least two from a smooth projective rationally connected manifold preserves the Oka property (Corollary 8.4).

A *holomorphic spray* on Y is a holomorphic map $s : E \to Y$ from a holomorphic vector bundle $E \to Y$ such that $s(0_y) = y$. It is *dominating* if $d(s|_{E_y})_{0_y}$ is surjective for every y; a manifold admitting such a spray is *holomorphically elliptic*. Gromov's theorem implies that every holomorphically elliptic manifold is Oka [40]. For smooth hypersurface complements we obtain:

Theorem 1.11. *Let $D \subset \mathbb{P}^n$ be a smooth hypersurface of degree $d \geq 1$, where $n \geq 1$. The following conditions are equivalent:*

(i) $\mathbb{P}^n \setminus D$ *is Oka;*
(ii) $\mathbb{P}^n \setminus D$ *is holomorphically elliptic;*
(iii) $d \leq n+1$.

For $n = 1$, smoothness means that D consists of d distinct points; the positive cases are $\mathbb{C}$ and $\mathbb{C}^*$.

For complements, Kusakabe treated the quadric case [44] and Xie the smooth cubic case [56]; the plane cubic case answered Problem B of Forstnerič and Lárusson [28]. Theorem 1.11 resolves Forstnerič's Problem 5.3(a) for smooth hypersurfaces [37] and settles the complement part of Problem E of Forstnerič and Lárusson [28] for every smooth hypersurface, with the exact threshold $d \leq n+1$. At the critical degree $d = n+1$, successive polar equations produce families of lines meeting D in at most two distinct points, whose complements are parametrized by entire curves. For $d \geq n+2$, the rank obstruction of Carlson and Griffiths [13] excludes the Oka property.

Relation to the work of OpenAI. Our proof that LAP implies the Oka property builds on the local-to-global strategy of OpenAI [49]: local planar extension with passive holomorphic parameters leads, through polynomial source coordinates, to polydisc approximation and then to convex approximation by removing real faces. The concepts of transverse complete directions, expanding finite strips, and thin transverse source coordinates are central inspirations. We also adapt the center-gluing and boundary-disc propagation arguments used in that framework. The formulation of LAP is also inspired by the local extension condition introduced in [49]; Appendix A proves their pointwise equivalence.

The contributions of the present authors are the formulation of LAP in arbitrary dimension, the proof that it implies the Oka property, and the analytic refinements that make this criterion effective. These include exact fiber-coordinate normalization in scalar substitution, an explicit planar corridor collapse with simultaneous sector gluing, and a near-identity estimate for normalized inverse iterates that is uniform in the number of iterates. For the convex step, one coarse entire spray fixes a complex collar; polynomial approximation of its local fiber inverse then gives the accuracy needed for ordinary bounded gluing. For the geometric applications, we combine this criterion with the classical and recent results cited in their proofs.

How to read the proof. The proof begins by using centered families to extend arbitrary maps with image in a small target neighborhood: scalar substitution and exact partial compositions provide this local operation. Polynomial source coordinates then give thin boundary fibers, and planar enlargement with transverse and spray variables retained as passive parameters gives polydisc approximation. A fixed annular overlap joins the enlarged family to the old map. Induction across real faces finally yields CAP.

The paper follows this progression. Section 2 treats centered entire families, graph coordinates, and center gluing. Section 3 develops scalar substitution and local planar extension; Sections 4–6 pass through polynomial source geometry and polydisc approximation to convex approximation. Subsections 7.1 and 7.2 treat small exceptional sets, holomorphic images, complements, and group orbits. Sections 8 and 9 give the geometric applications, and Section 10 concludes with three conjectures and their compact Kähler analogues.

2. Local geometry and gluing

2.1. **The geometric condition behind strong dominability.** Let Y be a complex manifold of dimension $n \geq 1$. We use LAP and its locus G_Y from Definition 1.2.

Lemma 2.1. *Let $V \subset Y$ be an open neighborhood of x. Suppose $h_j : V \times \mathbb{C} \to Y$, $1 \le j \le n$, are holomorphic, $h_j(y,0) = y$, and the vectors $\partial_t h_j(x,0)$ form a basis of T_xY. Then $x \in G_Y$.*

Proof. Immediate from the definition. □

Proposition 2.2. *Every strongly dominable complex manifold satisfies* LAP. *More precisely, a dominating entire map at x supplies centered entire families near x whose initial velocities form a basis.*

Proof. Choose a holomorphic map $\Psi : \mathbb{C}^n \to Y$ and a point $a \in \mathbb{C}^n$ such that $\Psi(a) = x$ and $d\Psi_a$ is an isomorphism. Let $\lambda : V \to \mathbb{C}^n$ be its local inverse on a neighborhood of x. For the standard coordinate vectors e_j, set

$$h_j(y,t) = \Psi(\lambda(y) + te_j), \qquad (y,t) \in V \times \mathbb{C}.$$

These families are centered and their initial velocities at x are $d\Psi_a(e_j)$, a basis. If dominability is stated using a map from $\mathbb{C}^N$ with surjective differential, restrict it first to an affine n-plane on which that differential is an isomorphism. □

2.2. **Local coordinates and one-step gluing.** To glue maps defined on two overlapping source domains, we allow each map to vary in a small holomorphic family. The additional variable v parametrizes small changes of the original map, which corresponds to $v = 0$. We then seek small holomorphic choices $v = a(z)$ and $v = b(z)$ on the two domains so that the resulting maps agree on the overlap. The lemmas below construct these coordinates, split a holomorphic error into contributions from the two sides, and solve for a and b by a contraction. The output is one glued map.

2.2.1. *Conventions and graph coordinates.* Throughout, Y is a complex manifold of dimension n. We fix a complete distance d_Y inducing its topology; such a distance exists by the theorem of Nomizu and Ozeki [48, Theorem 1], applied componentwise. Completeness is used only when taking an exhaustion limit. We abbreviate $\Delta = \Delta_1$ and use the maximum norm $\|v\|_\infty = \max_j |v_j|$ to write $\Delta_r^q = \{v \in \mathbb{C}^q : \|v\|_\infty < r\}$. The notation $\mathcal{O}(\operatorname{Op} K, Y)$ means maps holomorphic on some neighborhood of K. A closed polydisc of dimension zero is a point.

For a holomorphic map $f : U \to Y$ on a complex manifold U and $b > 0$, a *chart spray* centered at f is a holomorphic family $F : U \times \Delta_b^n \to Y$ with $F(z,0) = f(z)$ for which $(z,v) \mapsto (z, F(z,v))$ is biholomorphic onto its image. The following standard graph construction is used by OpenAI [49, Lemma 2.6].

Lemma 2.3. *Let S be Stein, $f : S \to Y$ holomorphic, and suppose f^*TY is holomorphically trivial. For every $K \Subset S$ there are a neighborhood U of K, a number $b > 0$, and a chart spray $F : U \times \Delta_b^n \to Y$ centered at f.*

Proof. The graph of f is a Stein submanifold of $S \times Y$ and has a Stein neighborhood by Siu's theorem [53, Main Theorem]. A frame of its vertical tangent bundle extends, after shrinking that neighborhood, to holomorphic vertical vector fields: apply Cartan's theorem B to the restriction sequence for the vertical tangent sheaf. Compose their local flows. Near the zero section this gives a fiber-preserving holomorphic map whose vertical differential is the chosen frame. The inverse function theorem and compactness of K give one neighborhood U and one fiber radius on which the graph map is biholomorphic. □

In the applications, the source is a contractible Stein domain or a planar domain times a polydisc. A planar domain has the homotopy type of a one-dimensional CW complex, so complex vector bundles on it are topologically trivial. The Oka–Grauert principle [39, Satz I] therefore provides the triviality required in Lemma 2.3. When using the inverse of a chart spray F, we restrict the source to a compact subset $K_0 \Subset U$ and the fiber variable to $\overline{\Delta}_{b'}^n$ with $b' < b$. The image of $K_0 \times \overline{\Delta}_{b'}^n$ under $(z,v) \mapsto (z, F(z,v))$ is then a compact subset of the

open chart image. Consequently, sufficiently small perturbations of these values still lie in that image, so the inverse remains defined uniformly for $z \in K_0$.

2.2.2. *Bounded additive splitting.* For an open set $U \subset \mathbb{C}^m$, write $H^\infty(U, \mathbb{C}^n)$ for the Banach space of bounded holomorphic maps $u : U \to \mathbb{C}^n$, with

$$\|u\|_U = \sup_{z \in U} \max_{1 \le j \le n} |u_j(z)|.$$

Derivatives are measured in the operator norm induced by the maximum norm. We use pairs of bounded open sets $A, B \subset \mathbb{C}^m$ whose closed fringes $\overline{A \setminus B}$ and $\overline{B \setminus A}$ are disjoint. Set $C = A \cap B$.

Lemma 2.4. *Suppose $A \cup B$ is a smoothly bounded strongly pseudoconvex domain. There are a constant $M > 0$ and bounded linear operators*

$$R_A : H^\infty(C, \mathbb{C}^n) \to H^\infty(A, \mathbb{C}^n), \qquad R_B : H^\infty(C, \mathbb{C}^n) \to H^\infty(B, \mathbb{C}^n)$$

with

$$R_A u - R_B u = u, \qquad \|R_A u\|_A + \|R_B u\|_B \le M \|u\|_C.$$

For bounded open sets $A, B \subset \mathbb{C}$ with the same separation condition, the conclusion also holds on $A \times P, B \times P$ for every open polydisc P; the operators preserve holomorphic dependence on P, and their bound depends only on A, B.

Proof. The first assertion is the classical additive Cousin splitting in [21, Lemma 4.6]. For completeness, set $\Omega = A \cup B$ and choose a smooth function $\chi : \mathbb{C}^m \to [0, 1]$ with compact support, equal to 0 near $\overline{A \setminus B}$ and to 1 near $\overline{B \setminus A}$. For $u \in H^\infty(C, \mathbb{C}^n)$, the functions χu and $(\chi - 1)u$ extend smoothly by zero to A and B, respectively. Their $\bar\partial$ derivatives agree with the bounded, $\bar\partial$-closed form $\eta = u\bar\partial\chi$, extended by zero from C to Ω. A bounded linear $\bar\partial$ solution operator on Ω gives v with $\bar\partial v = \eta$. Hence

$$R_A u = \chi u - v|_A, \qquad R_B u = (\chi - 1)u - v|_B$$

are holomorphic, differ by u on C, and satisfy the required bound; the products denote their zero extensions.

For the planar assertion, use the Cauchy–Green operator on the coefficient $u\bar\partial\chi$, extended by zero to the plane; see [36, Lemma 3.2 and the proof of Lemma 3.3]. The kernel estimate gives a bound depending only on the fixed planar sets. Since the cutoff and integration involve only the planar variable, the operators preserve holomorphic dependence on the parameters in P, with the same bound for every P and its dimension. Apply the construction componentwise. □

2.2.3. *Only the centers need to be glued.* The following form of OpenAI's center-gluing argument [49, Lemma 2.8] is used throughout the paper. It separates the geometry, which determines the splitting bound M, from the accuracy of the families.

Lemma 2.5. *Suppose the operators of Lemma 2.4 are fixed. Let F_A be a holomorphic chart spray on $A \times \Delta^n_{\rho_+}$ and F_B a holomorphic family on $B \times \Delta^n_\rho$, where $\rho_+ > \rho > 0$. Suppose their transition is defined on $C \times \Delta^n_\rho$ and has the form*

$$F_A(z, \cdot)^{-1}(F_B(z, v)) = v + c(z, v).$$

Set $e = \|c\|_\infty$ and $\kappa = \|D_v c\|_\infty$ there. If $Me < \rho/2$ and $M\kappa < 1/2$, there are holomorphic corrections a on A and b on B satisfying

$$F_A(z, a(z)) = F_B(z, b(z)) \quad (z \in C), \qquad \|a\|_A + \|b\|_B \le Me.$$

The same conclusion holds if the families also depend holomorphically on $p \in P$, provided the splitting operators act on functions of (z, p), preserve holomorphic dependence on p, and have the same bound M uniformly on P. Here the radii are independent of p, and e and κ are the supremum norms on $C \times P \times \Delta^n_\rho$, subject to the same inequalities. The corrections are then

holomorphic functions $a(z,p)$ and $b(z,p)$, and the estimates are uniform in p. Lemma 2.4 supplies such operators for planar A, B.

Proof. On the closed ball $\|u\|_C \le e$ in $H^\infty(C,\mathbb{C}^n)$, set

$$\mathcal{T}(u)(z) = c(z, R_B u(z)).$$

The first inequality keeps $R_B u$ in the prescribed fiber domain. We have $\|\mathcal{T}(u)\|_C \le e$ and $\|\mathcal{T}(u) - \mathcal{T}(\widetilde{u})\|_C \le M\kappa\|u - \widetilde{u}\|_C$. At the fixed point set $a = R_A u$, $b = R_B u$. Then $a - b = u = c(z,b)$, which is exactly the transition identity required for the centers to agree. The norm estimate follows from the linear splitting bound. □

Consequently the glued map changes the old center by an amount tending to zero with e, uniformly on every retained compact set. This is enough to iterate: in our applications the enlarged compact has a contractible Stein neighborhood inside the domain of the glued map. Its pulled-back tangent bundle is therefore trivial, and Lemma 2.3 constructs a new chart spray there for the next step, so the gluing proceeds without carrying a spray across the overlap. Inverse fiber charts are required only for the old family. If the transition is uniformly small on a larger fiber polydisc, its derivative is small on a smaller one by Cauchy estimates; this is how the hypothesis on κ will always be obtained.

3. From disc families to local extension

We first work in one complex source coordinate, with the remaining coordinates serving as holomorphic parameters. The goal is to extend approximation from an old disc to a larger disc by gluing along a thin overlap. This is the gluing idea underlying OpenAI's two-strip argument [49, Proposition 2.2], adapted below to centered disc families.

Definition 3.1. A pair $D \subset D' \subset \mathbb{C}$ is a *special planar pair* [49, Section 2, before Definition 2.1] if D and D' are closed Jordan discs with piecewise smooth boundaries and there is a closed Jordan disc $E \subset \mathbb{C}$ with piecewise smooth boundary such that $D' = D \cup E$ and $D \cap E$ is homeomorphic to $[0,1]$ and is a proper subset of both ∂D and ∂E.

We use slightly larger neighborhoods when making source splittings.

The model to keep in mind is a centered family $h_\nu : V \times \Delta_{R_\nu} \to Y$, with $h_\nu(y,0) = y$. Here s is the source point, w allows a small change of the old map, and t is the scalar parameter to be replaced by a holomorphic function of s. If $G(s,w)$ is a small chart spray around an old map f_0, with image in V, then the families below are

$$T_\nu(s,w,t) = h_\nu(G(s,w),t).$$

The approximation class in the lemma records which maps on the old product can already be approximated from the enlarged product. The lemma says that substitution into the limiting family preserves this approximation property.

Lemma 3.2. *Let $D \subset D'$ be a special planar pair and, for an integer $q \ge 0$ and a closed polydisc $P \subset \mathbb{C}^q$, set $K = D \times P$ and $L = D' \times P$. The class $\mathcal{A}$ consists of maps to Y holomorphic near K that are uniformly approximable on K by maps to Y holomorphic near L.*

Suppose $f_0 \in \mathcal{A}$, $\dim Y = n$, and

$$G : \Omega \times \Delta_r^n \longrightarrow Y, \qquad G(s,0) = f_0(s),$$

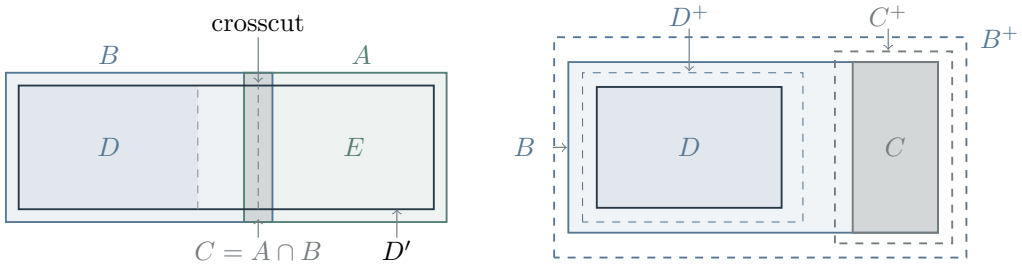
is a chart spray on a fixed neighborhood Ω of K; that is, $(s,w) \mapsto (s, G(s,w))$ is biholomorphic onto its image. Suppose the holomorphic maps

$$T_\nu : \Omega \times \Delta_r^n \times \Delta_{R_\nu} \longrightarrow Y, \qquad T_\nu(s,w,0) = G(s,w), \qquad R_\nu \longrightarrow \infty,$$

converge locally uniformly on $\Omega \times \Delta_r^n \times \Delta_\tau$ to T, for some $\tau > 0$. If φ is holomorphic near K and $\|\varphi\|_K < \tau$, then $T(\cdot, 0, \varphi) \in \mathcal{A}$.

We begin with a map g on the larger product that approximates f_0 on the old one. Figure 1 shows the two sides used for gluing: B contains the old disc D, while A covers the rest of the enlarged disc, and their overlap C is disjoint from D. The family on A remains centered at g; on B, a scalar polynomial selects a deformation by approximating φ near D and zero near C. An exact change of spray coordinates makes the two families agree at scalar zero on the whole fiber. They are therefore close on the overlap and can be glued with small corrections, using the standard spray-gluing method in the proof of [23, Proposition 3.1].

The order of choices is essential. The domains and gluing tolerance are fixed first, then the scalar polynomial, and finally the family index ν. Away from D and C, the polynomial may be large; its values need only lie inside Δ_{R_ν} there. On the overlap, convergence is used for scalar values close to zero; on the old product, it is used on a fixed scalar disc containing $\varphi(K)$ and compactly contained in Δ_τ, with the potentially large values elsewhere controlled only by the radius condition. The proof below carries out the normalization, scalar approximation, and gluing in this order.



(a) The two gluing sides (b) Neighborhoods on the old side

FIGURE 1. The planar sets in the proof of Lemma 3.2, shown schematically. In (a), the thickened sides cover D' and their intersection C is disjoint from D. In (b), the disjoint neighborhoods D^+ and C^+ lie inside B^+, which also contains B. The passive polydisc factor is omitted.

Proof. Step 1: fix the old approximant, domains, and normalization. Fix $\epsilon > 0$ and $\|\varphi\|_K < t_1 < \tau$. Choose a map g holomorphic near L, sufficiently close to f_0 on K that

$$\rho(s) = G(s,\cdot)^{-1}(g(s))$$

is holomorphic near K, has values compactly inside Δ_r^n, and

$$\sup_K d_Y\big(T(s,\rho(s),\varphi(s)), T(s,0,\varphi(s))\big) < \epsilon/3.$$

This follows from compactness and continuity of the inverse chart and of T.

A crosscut in the attached disc, close to but disjoint from D, gives a thin closed collar B containing D and a complementary closed region A, with slightly thickened boundaries, such that $D' \subset \operatorname{Int}(A \cup B)$ and the overlap $C = A \cap B$ is a closed disc disjoint from D; see Figure 1(a). Take the collar sufficiently thin that the original G-chart and ρ are available near its product with P. Choose disjoint closed disc neighborhoods D^+, C^+ of D, C, a slightly larger closed neighborhood B^+ of B containing them, as in Figure 1(b), and fixed polydiscs in the remaining q source coordinates

$$P \Subset \operatorname{Int} P^+ \Subset \operatorname{Int} P^{++}.$$

Here Int denotes topological interior. These coordinates are *passive* only in the geometric construction: we enlarge the planar disc while keeping these polydiscs fixed. The maps may still depend holomorphically on all source coordinates. Make these choices inside the relevant holomorphic domains, with $B^+ \times P^{++} \Subset \Omega$, with g in the image of the G-chart there, and

with φ holomorphic near $D^+ \times P^+$. Fix separated open gluing domains U_A, U_B and an open polydisc U_P satisfying

$$D' \subset U_A \cup U_B, \quad P \Subset U_P \Subset \operatorname{Int} P^+, \quad U_A \cap U_B \Subset \operatorname{Int} C^+, \quad U_B \Subset \operatorname{Int} B^+.$$

Their planar additive-splitting constant is now fixed.

Take a chart spray $F(s,v)$ centered at g over a neighborhood of the two sides. Lemma 2.3 applies on a sufficiently small contractible Stein neighborhood of the disc product. Choose a fixed $b > 0$ so that F is available for $v \in \Delta^n_{4b}$ and

$$W(s,v) = G(s,\cdot)^{-1}(F(s,v))$$

is holomorphic near $B^+ \times P^{++} \times \overline{\Delta}^n_{3b}$, with image compactly inside Δ^n_r. Here $W(s,0) = \rho(s)$. Set

$$Q_\nu(s,v,t) = T_\nu(s, W(s,v), t), \qquad Q(s,v,t) = T(s, W(s,v), t).$$

These families converge locally uniformly for $|t| < \tau$, and, for every allowed fiber value,

$$Q_\nu(s,v,0) = Q(s,v,0) = F(s,v). \tag{3.1}$$

Normalization on the whole fiber, rather than only at its center, makes the overlap transition for the limiting family tend to the identity as the scalar parameter tends to zero. The fixed fiber margin then gives derivative control by Cauchy estimates.

Step 2: choose the accuracies, then the scalar polynomial. By uniform continuity of Q, choose $0 < \alpha < b$ and $\xi > 0$ so that, for $s \in K$, $\|v\| \le \alpha$, and $|u - \varphi(s)| < \xi$, one has $|u| < t_1$ and

$$d_Y\big(Q(s,v,u), Q(s,0,\varphi(s))\big) < \epsilon/3.$$

Choose a transition tolerance $e > 0$ so small that a transition within e of the identity on the fixed overlap and Δ^n_{2b} admits zero-section gluing corrections of norm less than α. This is Lemma 2.5; its derivative hypothesis on Δ^n_b follows from Cauchy estimates on the larger fiber polydisc.

Let $\Psi_s = F(s,\cdot)^{-1}$. By (3.1), compactness, and the inverse-chart margins, choose $0 < \eta < \tau$ such that $\Psi_s(Q(s,v,t))$ is defined and within $e/2$ of v on $C^+ \times P^+ \times \overline{\Delta}^n_{2b}$ whenever $|t| \le \eta$. Local uniform convergence gives the same assertion with Q_ν and tolerance e for every sufficiently large ν, on this same fixed compact product.

The compact set $(D^+ \cup C^+) \times P^+$ is polynomially convex. Oka–Weil approximation [25, §2.3] therefore supplies a polynomial $\psi : \mathbb{C}^{1+q} \to \mathbb{C}$ such that

$$\|\psi - \varphi\|_{D^+ \times P^+} < \xi, \qquad \|\psi\|_{C^+ \times P^+} < \eta. \tag{3.2}$$

Step 3: choose the family index last, then glue. Fix ψ and choose ν so large that the preceding overlap estimates hold,

$$R_\nu > 1 + \|\psi\|_{B^+ \times P^+}, \tag{3.3}$$

and

$$\sup_{K \times \overline{\Delta}^n_b \times \overline{\Delta}_{t_1}} d_Y(Q_\nu, Q) < \epsilon/3.$$

The convergence estimate is on a compact product fixed before ψ. Its possibly large values elsewhere on $B^+ \times P^+$ enter only the radius requirement (3.3).

Use the two families

$$F_A(s,v) = F(s,v), \qquad F_B(s,v) = Q_\nu(s,v,\psi(s)).$$

The second is defined throughout the fixed B side by (3.3). On the overlap its transition into the F_A-chart is within e of the identity by (3.2). Lemma 2.5 provides holomorphic corrections a and b_*, both of norm less than α, with

$$F_A(s, a(s)) = F_B(s, b_*(s)).$$

Thus the two expressions define $\widetilde{g}$ holomorphic on $(U_A \cup U_B) \times U_P$, a neighborhood of L. On K,

$$\widetilde{g}(s) = Q_\nu(s, b_*(s), \psi(s)).$$

The three error terms below are each less than $\epsilon/3$ by the choices above:

$$\begin{aligned} &d_Y\big(\widetilde{g}(s), T(s,0,\varphi(s))\big) \\ &\quad \le d_Y\big(Q_\nu(s,b_*(s),\psi(s)), Q(s,b_*(s),\psi(s))\big) \\ &\qquad + d_Y\big(Q(s,b_*(s),\psi(s)), Q(s,0,\varphi(s))\big) \\ &\qquad + d_Y\big(T(s,\rho(s),\varphi(s)), T(s,0,\varphi(s))\big) < \epsilon. \end{aligned}$$

□

Remark 3.3. For an actual centered family $h : V \times \mathbb{C} \to Y$, take $T_\nu(s,w,t) = h(G(s,w),t)$ on increasing scalar discs. The lemma then permits every scalar function holomorphic near K: choose $\tau > \|\varphi\|_K$. The family F_B need neither dominate nor remain in the old fiber chart away from the overlap.

3.1. **Local extension by exact partial compositions.**

Lemma 3.4. *For every x in the* LAP *locus G_Y, there is an open neighborhood $W_x \ni x$ with the following property: if $D \subset D'$ is a special planar pair, $P \subset \mathbb{C}^q$ is a closed polydisc, and k is holomorphic near $D \times P$ with values in W_x, then k is uniformly approximable on $D \times P$ by maps holomorphic near $D' \times P$ with values in Y.*

Proof. For $a = (a_1, \dots, a_n)$ in a sufficiently small polydisc $Q \subset \mathbb{C}^n$ about 0, use the families of Definition 1.2 to form the exact partial compositions

$$\theta_0(a) = x, \qquad \theta_j(a) = h_j(\theta_{j-1}(a), a_j), \qquad \Theta(a) = \theta_n(a).$$

Shrink Q so that its closure lies in their domains, every partial image is relatively compact in V, and $|a_j| < \tau$ on $\overline{Q}$. Since $h_j(y,0) = y$, one has $d_y h_j(y,0) = \mathrm{id}$. The columns of $d\Theta_0$ are therefore $\partial_t h_j(x,0)$. The inverse function theorem gives a neighborhood W_x whose closure lies in an inverse chart for Θ, with $\Theta^{-1}(\overline{W}_x) \Subset Q$.

For the given k, write $a = \Theta^{-1} \circ k$ and set

$$k_0 = x, \qquad k_j = h_j(k_{j-1}, a_j), \qquad k_n = k.$$

Let $\mathcal{A}$ be the approximation class in Lemma 3.2 for this fixed pair and this fixed P. The constant map k_0 belongs to $\mathcal{A}$. Suppose $k_{j-1} \in \mathcal{A}$. By Lemma 2.3, choose a chart spray $H_j(s,w)$ centered at k_{j-1}. Compactness permits fixed base and fiber neighborhoods on which its image lies in V. On these neighborhoods set

$$T_{j,\nu}(s,w,t) = h_{j,\nu}(H_j(s,w),t).$$

The families have exact normalization $T_{j,\nu}(s,w,0) = H_j(s,w)$ and converge locally uniformly for $|t| < \tau$. Since $\|a_j\|_{D\times P} < \tau$, Lemma 3.2 gives $k_j \in \mathcal{A}$. Induction proves the result. □

4. Polynomial source geometry

Rosay–Rudin's attracting-basin construction [51, Section 9] obtains coordinates by compensating holomorphic iterates with inverse iterates of the linear part. The maps below are scaled compositions of automorphisms of the shift-like type studied by Bedford and Pambuccian [6, Section 1]. The finite-iterate source construction and its thin-fiber geometry are adapted from OpenAI [49, Lemma 3.3 and the proof of Proposition 3.4]. We give the estimates needed for polydisc extension, including a bound for the normalized inverse iterates that is uniform in the number of iterates.

The source construction has two geometric goals: to move points only slightly on the old polydisc, and to provide coordinates on the outer region in which each transverse slice over a scalar boundary point has small diameter. Figure 2 illustrates these two features. The

first preserves the approximation already obtained. The second ensures that, after applying the given map and allowing a small spray variation, each such slice lies in a single target neighborhood where the local extension lemma applies.

The roles of two parameters deserve emphasis. The polynomial degree controls the near-identity error and the thickness of the boundary slices, while the number of iterates allows us to reach any prescribed outer radius. The estimates are uniform in the number of iterates, so these parameters can be chosen in that order.

Lemma 4.1. *Fix $m \geq 2$ and numbers*

$$1 < r_m < \cdots < r_2 < r_1 < c < c_+ < b_0, \qquad 0 < a < 1,$$
$$r_j < a r_{j-1} \quad (2 \leq j \leq m).$$

For all sufficiently large integers d, set

$$P_j(w) = (w/r_j)^d, \qquad L(x) = (a x_m, -a x_1, \ldots, -a x_{m-1}),$$
$$H_d(x) = (a x_m, P_2(x_2) - a x_1, \ldots, P_m(x_m) - a x_{m-1}).$$

There are constants $C \geq 1$ and $B > 0$, independent of d and J, with the following properties: the map H_d is a polynomial automorphism. For every $J \geq 1$, let

$$E_{d,J} = H_d^{-J}(\overline{\Delta}_1^m), \qquad h_{d,J} = H_d^{-J} \circ L^J.$$

Then:

(1) On $E_{d,J}$ one has

$$|P_j(x_j)| \leq C \max(1, |x_1|) \quad (2 \leq j \leq m). \tag{4.1}$$

In particular, if $|x_1| \leq c_+$, then $x \in \Delta_{b_0}^m$.

(2) On a neighborhood of $E_{d,J} \cap \{|x_1| > r_1\}$ there are single-valued holomorphic functions $s, \xi_2, \ldots, \xi_m$ such that $|\xi_j| \leq B$ on that set and

$$x = \Phi_d(s, \xi), \qquad \Phi_{d,j}(s, \xi) = \rho_j s^{d^{m-j}} \exp\left(\sum_{k=2}^{j} \frac{\xi_k}{d^{j-k+1}} \right), \tag{4.2}$$
$$\rho_1 = 1, \quad \rho_j = r_j (a \rho_{j-1})^{1/d}.$$

The sum for $j = 1$ is empty; thus $x_1 = s^{D_d}$, where $D_d = d^{m-1}$. The map Φ_d is entire in (s, ξ).

(3) Set $\Xi_i = \{\xi \in \mathbb{C}^{m-1} : |\xi_j| \leq B + i\}$ for $i = 1, 2$, and $r = c^{1/D_d}$. Then

$$\Phi_d(\overline{\Delta}_r \times \Xi_2) \Subset \Delta_{b_0}^m, \qquad \sup_{|s|=r} \operatorname{diam}_\infty \Phi_d(\{s\} \times \Xi_2) = O(d^{-1}).$$

The constants in these assertions do not depend on J.

(4) For any fixed $1 < u_0 < u_1 < r_m$, set $\theta = u_1/r_m$ and $\alpha_d = a + \theta^d/u_1$. For sufficiently large d, simultaneously for every $J \geq 1$,

$$\|h_{d,J} - \mathrm{id}\,\|_{\overline{\Delta}_{u_0}^m} \leq \frac{\theta^d}{a - \alpha_d^d} \leq \frac{2}{a}\theta^d. \tag{4.3}$$

Consequently, for any $R_+ > 1$, one may first choose d to obtain any prescribed near-identity accuracy and thinness, and then choose J with $a^J R_+ < 1$. The resulting $h = h_{d,J}$ satisfies

$$h(\overline{\Delta}_{R_+}^m) \subset H_d^{-J}(\Delta_1^m) \subset E_{d,J}.$$

In particular $|h_1| < c$ on $\overline{\Delta}_1^m$ when d is large.

All inclusions asserted on compact sets have a positive margin, so they persist on a sufficiently small open neighborhood of those sets.

The proof is completed at the end of this section, after Lemma 4.4.

In the extension argument, assertion (1) keeps the inner part of the transported source in the domain of the old map, while assertion (2) expresses its outer part in one scalar coordinate and passive variables. Assertion (3) places each boundary fiber in a target neighborhood where local planar extension applies. The resulting extension is pulled back and glued to the old map on an inner overlap; assertion (4) controls the change on the original compact set.

In dimension two, the outer relation and its parametrization are

$$(x_2/r_2)^d = ax_1e^{\xi_2}, \qquad \Phi_d(s,\xi_2) = (s^d, r_2a^{1/d}se^{\xi_2/d}).$$

Thus s runs along the end, while bounded changes in ξ_2 move points in a fixed boundary fiber by only $O(d^{-1})$. In higher dimension, successive relations of this form produce the weighted powers of s in (4.2).

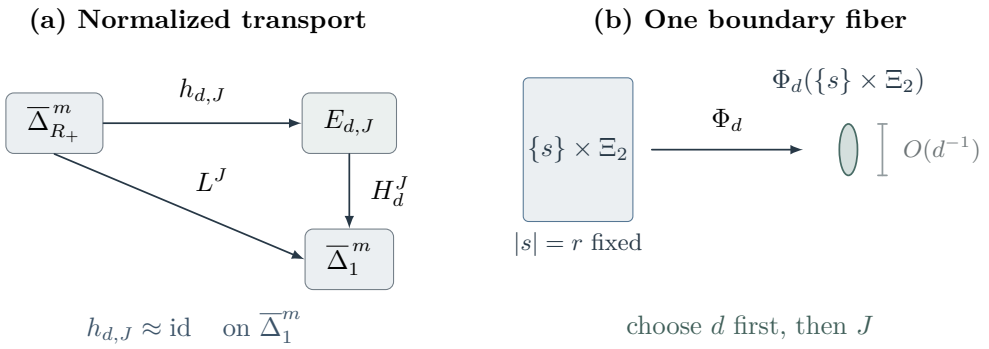


FIGURE 2. Two roles of the source construction, schematically. (a) Normalization gives $H_d^J \circ h_{d,J} = L^J$; the old polydisc is changed only slightly, while the prescribed larger source is sent into $E_{d,J}$. (b) At a fixed boundary value of s, the image of the transverse polydisc has maximum-norm diameter $O(d^{-1})$.

We use the fixed data and notation of Lemma 4.1. The three estimates below prove its power bound, outer coordinates, and uniform near-identity assertion. First, the inverse is obtained without solving any polynomial equation: if $y = H_d(x)$, set $x_m = y_1/a$ and then, successively,

$$x_{j-1} = (P_j(x_j) - y_j)/a \quad (j = m, m-1, \dots, 2).$$

This proves that H_d and its inverse are polynomial.

Lemma 4.2. *There are $C \geq 1$ and $q \in (0,1)$, independent of d and J, such that, for all sufficiently large d, the power bound (4.1) holds on $E_{d,J}$. If $x \in E_{d,J}$ and $|x_1| > r_1$, every x_j is nonzero and*

$$|H_d(x)_j| \leq qa|x_{j-1}| \quad (2 \leq j \leq m).$$

Moreover, $|x_1| \leq c_+$ implies $x \in \Delta^m_{b_0}$.

Proof. Choose $S > \max(2, r_1)$. Increasing the degree threshold ensures $(M/r_1)^d \geq 4M$ whenever $M \geq S$. The region

$$\mathcal{E} = \{x : \max_{j\geq 2} |x_j| > \max(S, |x_1|)\}$$

is forward invariant: if $M = |x_k| = \max_{j\geq 2}|x_j|$, then $|H_d(x)_k| \geq 3M$, whereas $|H_d(x)_1| \leq aM$. A forward orbit ending in $\overline{\Delta}^m_1$ cannot enter $\mathcal{E}$. Take $C > \max(1, 4a, 16a^2r_1^2, 4S)$. If $A = \max_{j\geq 2}|P_j(x_j)| > C\max(1,|x_1|)$, then $|x_j| \leq r_1\sqrt{A}$ for $j \geq 2$. For an index k attaining A, the term $a|x_{k-1}|$ is at most $A/4$, whether $k = 2$ or $k > 2$. Thus

$$|H_d(x)_k| \geq 3A/4 > S, \qquad |H_d(x)_1| \leq ar_1\sqrt{A} \leq A/4.$$

The next iterate belongs to $\mathcal{E}$, a contradiction. This proves (4.1); the same bound holds at every subsequent iterate, including the endpoint. If $|x_1| \le c_+$, it gives $|x_j| \le r_j(Cc_+)^{1/d} < b_0$ for large d. Together with $|x_1| \le c_+ < b_0$, this proves the inner inclusion.

To obtain the domination estimate, write $y = H_d(x)$, where $x \in E_{d,J}$, $J \ge 1$, and $|x_1| \ge r_1$. The power bound at y gives

$$|y_j| \le r_j\{C\max(1, a|x_m|)\}^{1/d}, \qquad 2 \le j \le m. \tag{4.4}$$

If $J = 1$, this follows instead from $|y_j| \le 1 < r_j$. Set

$$e_j = d^{m-j}, \qquad D_d = e_1 = d^{m-1}, \qquad t = (|x_1|/r_1)^{1/D_d} \ge 1,$$

and fix $R_* > r_1$ and $A_* = C\max(1, aR_*)$. We claim that

$$|x_j| \le R_* t^{e_j} \qquad (2 \le j \le m). \tag{4.5}$$

Indeed, let

$$\Lambda = \max\left\{1, \max_{2\le j\le m}\left(\frac{|x_j|}{R_* t^{e_j}}\right)^{1/e_j}\right\}.$$

Then $|x_j| \le R_*(\Lambda t)^{e_j}$ for every j, including $j = 1$, since $|x_1| = r_1 t^{D_d}$. If $\Lambda > 1$, equality holds at some $k \ge 2$. Using $P_k(x_k) = y_k + ax_{k-1}$, $e_{k-1} = de_k$, (4.4), and the bound $\max(1, a|x_m|) \le \max(1, aR_*)\Lambda t$, we obtain

$$\begin{aligned}(R_*/r_k)^d(\Lambda t)^{de_k} &\le aR_*(\Lambda t)^{de_k} + r_k A_*^{1/d}(\Lambda t)^{1/d},\\ (R_*/r_k)^d &\le aR_* + r_k A_*^{1/d}(\Lambda t)^{1/d-de_k} \le aR_* + r_k A_*^{1/d}.\end{aligned}$$

For all sufficiently large d this is impossible, simultaneously for $2 \le k \le m$, because $R_*/r_k > 1$. Hence $\Lambda = 1$, proving (4.5). In particular, $|x_m| \le R_* t$, so

$$|y_j| \le r_j A_*^{1/d} t^{1/d} \qquad (2 \le j \le m). \tag{4.6}$$

Choose $\delta > 0$ so small that $r_j - \delta > 0$ and

$$q = \max_{2\le j\le m}\frac{r_j + \delta}{a(r_{j-1} - \delta)} < 1.$$

For sufficiently large d, simultaneously for all $j \ge 2$,

$$r_j A_*^{1/d} \le r_j + \delta, \qquad r_j\{(1-q)a(r_{j-1} - \delta)\}^{1/d} \ge r_j - \delta.$$

Starting from $|x_1| = r_1 t^{D_d}$, induction gives

$$|x_j| \ge (r_j - \delta)t^{e_j} \quad (1 \le j \le m), \qquad \frac{|y_j|}{a|x_{j-1}|} \le q \quad (2 \le j \le m).$$

For if the lower bound holds at $j-1$, then (4.6) implies

$$\frac{|y_j|}{a|x_{j-1}|} \le q\, t^{1/d-e_{j-1}} \le q.$$

Consequently

$$|P_j(x_j)| \ge (1-q)a|x_{j-1}| \ge (1-q)a(r_{j-1} - \delta)t^{e_{j-1}},$$

and taking the dth root proves the next lower bound. Thus every x_j is nonzero, and the required domination estimate follows. All degree thresholds are independent of J. □

Lemma 4.3. *Let $B = -\log(1-q)$, with q from Lemma 4.2. The coordinate representation and thin-fiber assertions in parts (2) and (3) of Lemma 4.1 hold.*

The escape estimate makes each power term a controlled perturbation of a times the preceding coordinate. Taking logarithms records this perturbation in bounded variables; their coefficients in the new coordinates decrease with the degree, yielding thin transverse fibers.

Proof. Write $y = H_d(x)$. The ratios $P_j(x_j)/(ax_{j-1}) = 1 + y_j/(ax_{j-1})$ belong to $\{|w-1| \le q\}$. The branch of the logarithm centered at 1 therefore defines ξ_j holomorphically on a neighborhood of the end, with $|\xi_j| \le B = -\log(1-q)$. Define

$$s = \frac{x_m}{\rho_m} \exp\left(-\sum_{k=2}^{m} \frac{\xi_k}{d^{m-k+1}} \right).$$

The identities $x_j^d = r_j^d a x_{j-1} e^{\xi_j}$ prove (4.2) backwards, beginning with $j = m$. This also proves $s^{D_d} = x_1$, without any choice of a root.

For the thin-fiber estimate, the recurrence defining ρ_j shows that $|\log \rho_j|$ is bounded independently of d, and $\log \rho_j - \log r_j = O(d^{-1})$ for $j \ge 2$. There is a constant $K > 0$, independent of d, such that for $|s|^{D_d} = c$, $\xi \in \Xi_2$, and $2 \le j \le m$, the logarithm of $|\Phi_{d,j}/r_j|$ is bounded above by K/d: the power of s contributes $\log c/d^{j-1}$ and the transverse sum has modulus at most $(B+2)/(d-1)$. Thus

$$|\Phi_{d,j}(s,\xi)| \le r_j e^{K/d} \quad (j \ge 2).$$

The corresponding transverse sums at two points of Ξ_2 differ by at most $2(B+2)/(d-1)$. Using $|e^u - e^v| \le |u-v| e^{\max(|u|,|v|)}$ proves the asserted $O(d^{-1})$ diameter bound. The first component is independent of ξ and has modulus $c < b_0$. Every component contains a positive integral power of s, so the boundary bounds hold on the full closed scalar disc as well. Since $r_j < b_0$, they imply the stated compact inclusion for large d. □

Lemma 4.4. *For each fixed $1 < u_0 < u_1 < r_m$, the estimate (4.3) holds for all sufficiently large d, uniformly over all $J \ge 1$.*

Proof. Write $H_d = L + N_d$. In the maximum norm, $\|L^k x\|_\infty = a^k \|x\|_\infty$ for every integer k and

$$\|N_d(x)\|_\infty \le (\|x\|_\infty / r_m)^d.$$

On $\overline{\Delta}_{u_1}^m$ this gives $\|H_d(x)\|_\infty \le \alpha_d \|x\|_\infty$. For large d, $\alpha_d < 1$ and $\alpha_d^d \le a/2$. All forward iterates stay in this polydisc and satisfy $\|H_d^k(x)\|_\infty \le \alpha_d^k u_1$. For $g_{d,J} = L^{-J} \circ H_d^J$, telescoping now gives

$$g_{d,J}(x) - x = \sum_{k=0}^{J-1} L^{-k-1} N_d(H_d^k(x)), \qquad \|g_{d,J} - \mathrm{id}\,\|_{\overline{\Delta}_{u_1}^m} \le \varepsilon_d := \frac{\theta^d}{a - \alpha_d^d}.$$

The global inverse of $g_{d,J}$ is $h_{d,J}$, and both fix zero. Increase d so that $\varepsilon_d < u_1 - u_0$. For $z \in \overline{\Delta}_{u_0}^m$, the path $h_{d,J}(tz)$, $0 \le t \le 1$, cannot first reach the boundary of $\Delta_{u_1}^m$: at that point the preceding estimate would give

$$\|tz\|_\infty = \|g_{d,J}(h_{d,J}(tz))\|_\infty \ge u_1 - \varepsilon_d > u_0.$$

Applying the estimate for $g_{d,J}$ at $h_{d,J}(z)$ proves (4.3). All degree thresholds are independent of J. □

Proof of Lemma 4.1. The inverse recurrence proves polynomial invertibility. Lemmas 4.2–4.4 give all four estimates, with one common degree threshold independent of J. Choose d beyond this threshold and with the required near-identity accuracy and fiber diameter. For the prescribed R_+, choose J so that $a^J R_+ < 1$. Then L^J sends $\overline{\Delta}_{R_+}^m$ into Δ_1^m, proving the outer-domain inclusion. The uniform convergence $h_{d,J} \to \mathrm{id}$ near $\overline{\Delta}_1^m$ also gives $|h_1| < c$ there. Strict inequalities on the compact sets in question persist on small neighborhoods, completing the proof. □

5. Extension across polydiscs

We now convert the local sector extension of Lemma 3.4 into approximation on polydiscs. The thin boundary fibers from Section 4 provide the target-neighborhood containment needed for this step.

Throughout this section, the neighborhoods W are those supplied by Lemma 3.4: a map near $J \times P$ with values in W admits approximation by maps near $L \times P$, for every special planar pair $J \subset L$ and prescribed closed passive polydisc P. The same P occurs in both products.

5.1. **The scalar collapse.** One entire function simultaneously preserves the old disc and brings all the attaching corridors back to its boundary.

Lemma 5.1. *For $r > 0$ and an integer $N \geq 1$, set*

$$p_N(s) = \int_0^s \exp(-(t/r)^N)\, dt, \qquad \zeta_j = re^{2\pi ij/N} \quad (0 \leq j \leq N).$$

Then

$$\sup_{|s|\leq r} |p_N(s) - s| \leq \frac{r(e-1)}{N+1}, \qquad \sup_{t\geq r} |p_N(te^{2\pi ij/N}) - \zeta_j| \leq \frac{r}{N+1}.$$

Proof. Termwise integration of the exponential series gives the first estimate. On the positive real ray,

$$0 \leq r - p_N(r) \leq \frac{r}{N+1}, \qquad \int_r^\infty e^{-(t/r)^N}\, dt \leq \frac{r}{eN} \leq \frac{r}{N+1}.$$

For the integral estimate substitute $u = (t/r)^N$ and use $u^{1/N-1} \leq 1$ for $u \geq 1$. Monotonicity gives the second estimate on this ray. The identity $p_N(\omega s) = \omega p_N(s)$ for $\omega^N = 1$, which follows from the power series, gives the remaining rays. □

5.2. **Disc enlargement with fixed passive parameters.** The planar model is OpenAI's spokes-and-sectors enlargement [49, Proposition 2.9]: boundary values are kept in fixed target neighborhoods while the sectors between radial spokes are filled. Here the scalar collapse constructs all corridors at once, and the sectors are filled simultaneously.

Lemma 5.2. *Let $r > 0$, P be a closed polydisc, and f holomorphic near $\overline{\Delta}_r \times P$. Suppose that, for every $|\zeta| = r$, the compact fiber $f(\{\zeta\} \times P)$ is contained in one of the neighborhoods W. For every $S > r$, f is uniformly approximable on $\overline{\Delta}_r \times P$ by maps holomorphic near $\overline{\Delta}_S \times P$.*

Proof. By Lemma 2.3, choose a chart spray $f^\sharp$ centered at f. Compactness permits a slightly larger passive polydisc P^+ and a fixed spray radius $\beta > 0$ such that finitely many boundary-arc products, including P^+ and $\overline{\Delta}^n_\beta$, map into their respective neighborhoods W with positive margins. Fix $0 < \rho < \rho_+ < \beta$.

Choose a sufficiently fine uniform subdivision of the circle and then p_N from Lemma 5.1. The family $f^\sharp(p_N(s), p, v)$ is defined on the old disc and on thin radial corridors extending beyond S, as in Figure 3(a). Its values near each boundary arc and the two adjacent corridors lie in the corresponding neighborhood W, uniformly for $(p, v) \in P^+ \times \overline{\Delta}^n_\beta$. This follows from the near-identity and collapsed-ray estimates, respectively.

Take the old side to be the inner disc together with the corridors, and the new side to be the disjoint intervening sectors, covering a neighborhood of $\overline{\Delta}_S$. Small boundary buffers within the controlled neighborhoods give separated closed fringes. For each sector, the overlap lies in the U-shaped collar J_j shown in Figure 3(b). Its bottom follows the boundary arc and its sides follow the two corridors, so the whole parameter family on J_j takes values in one W. Filling the U gives a sector L_j whose outer edge lies beyond the new side; the added disc meets J_j along a proper boundary arc, hence $J_j \subset L_j$ is a special planar pair. Compactness

allows all collars and buffers to be chosen uniformly in (p, v). Fix them before choosing the sector-approximation accuracy.

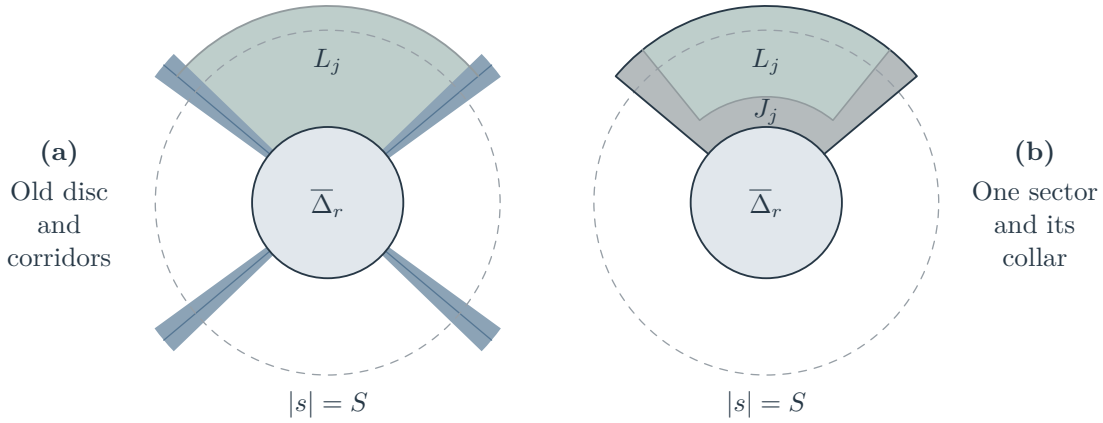


FIGURE 3. The planar construction in Lemma 5.2. (a) The old disc and radial corridors form the old side; the sectors are filled simultaneously. The sector L_j is shown in both panels. (b) Its collar J_j consists of bottom and side strips. The middle of the outer edge is not part of the collar. Widths are enlarged for visibility; small boundary buffers and the parameter factors are omitted.

Denote the scalar old side by A and the new side by $B = \bigsqcup_j B_j$, with $B_j \subset L_j$ and $A \cap B_j \subset J_j$. On A use the chart spray $F_A(s, p, v) = f^\sharp(p_N(s), p, v)$. Local sector extension, with (p, v) passive, supplies maps G_j near $L_j \times P^+ \times \overline{\Delta}^n_\beta$ arbitrarily close to F_A on the respective collars. Define $F_B = G_j$ on each B_j; the disjointness of the branches makes this a holomorphic family on the whole new side. Since there are finitely many sectors, their approximation errors can all satisfy one prescribed gluing tolerance. Apply the parameter version of Lemma 2.5 once to F_A, F_B on A, B, retaining p in a fixed neighborhood of P. The splitting operators act in the scalar variable and preserve holomorphic dependence on p. The fixed gap $\rho_+ - \rho$ supplies the required derivative smallness by Cauchy estimates. This gives a map holomorphic near $\overline{\Delta}_S \times P$, as close as desired to $f(p_N(s), p)$ on the old product. First making N large and then making the sector approximations accurate proves the assertion. □

5.3. **Finite extension.**

Proposition 5.3. *Suppose Y satisfies* LAP. *Let $m \geq 1$ and f be holomorphic near $Q = \overline{\Delta}^m$. For every $R > 1$ and $\epsilon > 0$, there is a map $\widetilde{f}$ holomorphic near $\overline{\Delta}^m_R$ such that*

$$\sup_Q d_Y(\widetilde{f}, f) < \epsilon.$$

Proof. When $m = 1$, choose a scalar radius $r > 1$ inside the domain of f. Each boundary value belongs to a neighborhood W, so Lemma 5.2, with no passive variables, gives the required finite extension.

Suppose $m \geq 2$. Choose $1 < b_0 < b' < b$ such that f is holomorphic near $\overline{\Delta}^m_b$, and a chart spray $f^\sharp(x, v)$ centered at f near $\overline{\Delta}^m_{b'} \times \{0\}$. For $R_+ > R$, take a smoothly bounded strongly convex domain D_0 with

$$\overline{\Delta}^m_R \Subset D_0 \Subset \Delta^m_{R_+}.$$

Choose the constants r_j, a, c, c_+ from Lemma 4.1, with $c_+ < b_0$, and choose

$$r_1 < c_0 < c_1 < c.$$

A finite cover of $f(\overline{\Delta}^m_{b_0})$ by the neighborhoods W has a positive Lebesgue margin. Choose a small common spray radius $\beta > 0$ and then a sufficiently large degree d in Lemma 4.1. Finally

choose J so that $a^J R_+ < 1$. Write $h = h_{d,J}$, $\Phi = \Phi_d$, $D_d = d^{m-1}$, and $r = c^{1/D_d}$. Thinness of the boundary fibers ensures that, for every $|s| = r$, the whole compact fiber

$$f^\sharp(\Phi(s,\xi),v), \qquad \xi \in \Xi_2, \quad v \in \overline{\Delta}_\beta^n,$$

lies in one neighborhood W. The degree can simultaneously be chosen so that

$$|h_1| < c_0 \quad \text{on } Q, \qquad \sup_Q d_Y(f\circ h, f) < \epsilon/2.$$

These choices are independent of the approximation accuracy below. If $|h_1| \le c_0$ throughout D_0, the inner inclusion in Lemma 4.1 makes $f\circ h$ holomorphic on D_0, and the assertion follows. Otherwise the outer side below is nonempty.

Here z denotes the original source point and $x = h(z)$ its transported image. The coordinates (s,ξ) describe only the outer part of that image; v is the spray variable. Scalar enlargement will extend the family in (s,ξ,v), after which it will be pulled back to the outer z-domain and glued to the old family.

The map

$$k(s,\xi,v) = f^\sharp(\Phi(s,\xi),v)$$

is holomorphic near $\overline{\Delta}_r \times \Xi_2 \times \overline{\Delta}_\beta^n$. The outer coordinates s,ξ are single-valued and holomorphic near $h(\overline{D}_0) \cap \{|x_1| \ge c_0\}$, and their actual transverse values lie strictly inside Ξ_1. Choose $S > r$ larger, with a positive margin, than all the scalar radii on this compact set. With (ξ,v) as passive variables, Lemma 5.2 gives maps $G(s,\xi,v)$ holomorphic near

$$\overline{\Delta}_S \times \Xi_1 \times \overline{\Delta}_\beta^n$$

and arbitrarily close to k on $\overline{\Delta}_r \times \Xi_1 \times \overline{\Delta}_\beta^n$.

Consider the fixed source pair

$$U_A = D_0 \cap \{|h_1| < c_1\}, \qquad U_B = D_0 \cap \{|h_1| > c_0\}.$$

It satisfies $U_A \cup U_B = D_0$, has separated closed fringes, and contains Q in $U_A \setminus \overline{U_B}$. On the two sides use

$$F_A(z,v) = f^\sharp(h(z),v), \qquad F_B(z,v) = G(s(h(z)),\xi(h(z)),v).$$

The first family is a chart spray near $\overline{U_A}$ by the inner inclusion in Lemma 4.1; the second is holomorphic near $\overline{U_B}$. On the overlap,

$$h(z) = \Phi(s(h(z)),\xi(h(z))), \qquad |s(h(z))| \le c_1^{1/D_d} < r.$$

Thus the pullback of k is exactly F_A on the overlap, while that of G is F_B. The approximation $G \approx k$ therefore makes F_B arbitrarily close to F_A on this fixed overlap and the full fiber product $\overline{\Delta}_\beta^n$.

Fix $0 < \rho < \rho_+ < \beta$. The source pair and its splitting constant are already fixed. The inverse charts of F_A and the Cauchy estimate across the fixed fiber gap make the transition and its fiber derivative as close to the identity as required by Lemma 2.5. That lemma gives a holomorphic map $\widetilde{f} : D_0 \to Y$ with

$$\sup_Q d_Y(\widetilde{f}, f\circ h) < \epsilon/2$$

when the scalar-enlargement approximation is sufficiently accurate. The two estimates give the desired approximation to f on Q. □

5.4. **Exhaustion.**

Theorem 5.4. *Suppose Y satisfies* LAP. *A map holomorphic near a closed polydisc in $\mathbb{C}^m$ can be approximated uniformly there by entire maps $\mathbb{C}^m \to Y$.*

Proof. Let f be the given map. Normalize its original polydisc to $Q = \overline{\Delta}^m$, fix $\epsilon > 0$, and set $f_0 = f$. Choose $1 = R_0 < R_1 < \cdots \to \infty$ and apply Proposition 5.3 successively after rescaling. We obtain maps f_j holomorphic near $\overline{\Delta}^m_{R_j}$ such that

$$\sup_{\overline{\Delta}^m_{R_{j-1}}} d_Y(f_j, f_{j-1}) < \epsilon 2^{-j-1}.$$

Completeness of d_Y gives a locally uniform limit $F : \mathbb{C}^m \to Y$. Target coordinate charts show that F is holomorphic, and the summable errors give the required approximation on Q. □

Remark 5.5. For a strongly dominable target, the sector-filling step in Lemma 5.2 has a direct form. Near each target point choose an entire map $\Psi : \mathbb{C}^n \to Y$ with an inverse branch λ on W. Let k_j be the old family on a collar product $J_j \times P$, with image in W and with all passive variables included in P. Its lift $\lambda \circ k_j$ is holomorphic near the polynomially convex product $J_j \times P$. Oka–Weil approximation by a polynomial q_j, followed by $\Psi \circ q_j$, fills the sector. The finite extension then follows from scalar enlargement and the fixed annular gluing above.

6. From polydisc approximation to convex approximation

The passage from polydiscs to arbitrary convex sets has two parts. First, approximation on lower-dimensional convex sets gives uniform approximation on compact convex subsets of a real hyperplane. Second, this real-face approximation allows us to remove one half-space constraint at a time. The second part requires a complex collar that remains fixed as the gluing accuracy improves; Subsection 6.3 explains how to obtain it.

The real-face induction follows OpenAI [49, Section 4]. There the lack of a prescribed complex collar is handled by gluing estimates on the real interface [49, Lemmas 4.4–4.5]; here the fixed-inverse approximation described in Subsection 6.3 supplies arbitrarily accurate transitions on one fixed collar. We use the approximation property with passive parameters from that argument, stated here in arbitrary target dimension.

In the following definition, r counts the convex source variables. The additional polydisc variables are retained throughout the induction: they include the auxiliary holomorphic parameters used to join local approximants and the spray variables needed for gluing.

Definition 6.1 (Convex approximation with passive parameters)**.** Let $r \geq 0$ be an integer. A complex manifold Y satisfies A_r if, for every integer $q \geq 0$, every compact convex set $K \subset \mathbb{C}^r$, every closed polydisc $P \subset \mathbb{C}^q$, every open neighborhood $U \subset \mathbb{C}^{r+q}$ of $K \times P$, every holomorphic map $f : U \to Y$, and every $\epsilon > 0$, there is a holomorphic map $E : \mathbb{C}^{r+q} \to Y$ such that

$$\sup_{K \times P} d_Y(E, f) < \epsilon.$$

A zero-dimensional factor is a point. In particular, A_0 is approximation on closed polydiscs in every dimension.

Theorem 6.2. *If Y satisfies A_0, then it satisfies A_r for every $r \geq 0$. In particular, Y has the convex approximation property.*

The proof is completed in Subsection 6.4, using the lemmas developed below.

We prove the implication $\mathsf{A}_{r-1} \Rightarrow \mathsf{A}_r$. All approximation assertions below concern uniform convergence; only fiberwise Cauchy estimates will be needed in the gluing step.

6.1. **Complexifying a real parameter.** We first replace a smooth real parameter by a holomorphic one. The complex neighborhood obtained in this step may depend on the desired accuracy, which is sufficient for the argument.

Lemma 6.3. *Let $I \subset \mathbb{R}$ be a compact interval and $C \subset \mathbb{C}^k$ compact and convex. Suppose that H is a smooth map to Y on a neighborhood of $I \times C$ in $\mathbb{R} \times \mathbb{C}^k$, holomorphic in the second variable for each real value of the first. For every $\epsilon > 0$ there is a map holomorphic on a complex neighborhood of $I \times C$ whose distance from H on $I \times C$ is less than ϵ.*

Proof. Choose a larger interval $I^+ = [a, b]$, a compact convex neighborhood C^+ of C, with all products under consideration in the domain of H, and a bounded convex open neighborhood X of C^+ with $I^+ \times X$ in that domain. The Stein-neighborhood theorem we need is the following specialization of [33, Corollary 2.2]: a continuous family of holomorphic sections over a Stein base, parametrized by a compact real interval, has a common Stein neighborhood of its graphs over any fixed holomorphically convex compact subset of the base. Here X is Stein, C^+ is $\mathcal{O}(X)$-convex because it is convex, and $q \mapsto (q, H(s, q))$ is such a family of sections of $X \times Y \to X$. The corollary gives a Stein neighborhood $\Theta \subset \mathbb{C} \times X \times Y$ of

$$\{(s, q, H(s, q)) : s \in I^+,\ q \in C^+\}.$$

The classical Stein embedding theorem [25] gives a closed holomorphic embedding $e : \Theta \hookrightarrow \mathbb{C}^N$. The holomorphic neighborhood-retraction theorem, in the form of [53, Corollary 1], gives a holomorphic retraction $\rho : \Omega \to e(\Theta)$ from an open neighborhood in $\mathbb{C}^N$.

Choose open neighborhoods U_0, U_1 of C with $C \Subset U_0 \Subset U_1 \Subset \operatorname{int} C^+$. The function $g(s, q) = e(s, q, H(s, q))$ is continuous as a map from I^+ into the Banach space $H^\infty(U_1, \mathbb{C}^N)$. Set $x = (s - a)/(b - a)$. The Bernstein polynomials of g

$$P_\nu(s, q) = \sum_{j=0}^{\nu} g\left(a + \frac{j(b-a)}{\nu}, q\right) \binom{\nu}{j} x^j (1 - x)^{\nu - j},$$

are holomorphic in (s, q) and converge uniformly to g on $I^+ \times U_1$. Indeed, uniform continuity controls the terms with $|j/\nu - x| \le \nu^{-1/4}$, while the variance bound $x(1 - x)/\nu \le 1/(4\nu)$ makes the total weight of the remaining terms at most $1/(4\sqrt{\nu})$.

For large ν, the compact set $P_\nu(I \times \overline{U}_0)$ lies in Ω. Hence it remains in Ω when I is replaced by a sufficiently small complex neighborhood V_ν. Let $\pi_Y : \mathbb{C} \times X \times Y \to Y$ be the projection. The maps

$$\pi_Y \circ e^{-1} \circ \rho \circ P_\nu : V_\nu \times U_0 \longrightarrow Y$$

approximate H uniformly on $I \times C$, by continuity on a compact neighborhood of the embedded graph. □

An interval contributes no new convex dimension: a planar coordinate change turns it into a passive disc, and scalar polynomial approximation removes that coordinate change afterward.

Lemma 6.4. *Assume A_s for some $s \ge 0$. If $I \subset \mathbb{R}$ is a compact interval, $C \subset \mathbb{C}^s$ is compact convex, and P is a closed polydisc, every map holomorphic near $I \times C \times P$ can be approximated uniformly there by maps entire in all displayed complex variables.*

Proof. For the given map H, choose a bounded simply connected complex neighborhood V of I such that H is holomorphic near $V \times C \times P$. A Riemann map $\varphi : V \to \Delta$ satisfies $\varphi(I) \Subset \Delta_\rho$ for some $\rho < 1$. Apply A_s to $H(\varphi^{-1}(\zeta), y, u)$ on $C \times (\overline{\Delta}_\rho \times P)$, regarding (ζ, u) as passive variables, and obtain an entire map $E(\zeta, y, u)$. Once E is fixed, approximate φ uniformly on I by a scalar polynomial p. The entire map $E(p(t), y, u)$ gives the assertion. Uniform continuity of the fixed E on a compact neighborhood of $\varphi(I) \times C \times P$ controls the substitution error. □

6.2. **Joining local approximants on a common domain.** We isolate the path construction in the proof of OpenAI [49, Lemma 4.3]. The local approximants along a real face will be entire, but the short path joining two nearby approximants is constructed in a chart of the original map. That path is initially defined only on a small joining product. To complexify the assembled family, we need all its paths on one common source polydisc. The next lemma obtains this by expanding the source near the entire endpoints, approximating the resulting family, and scaling back. The scaling gives endpoint control on the larger polydisc, where a small chart correction restores exact agreement.

Lemma 6.5. *Assume* A_s, $s \geq 0$. *Let* $C^-, C \subset \mathbb{C}^s$ *be compact convex sets with* $C^- \Subset \operatorname{int} C$, $B \subset \mathbb{C}^s$ *a closed polydisc containing* C *in its interior, and* $P \subset \mathbb{C}^q$ *a closed polydisc. Suppose that* $h_\tau(y,u)$, $0 \leq \tau \leq 1$, *is a smooth family holomorphic in* (y,u) *near* $C \times P$, *and that it agrees, on intervals near* $\tau = 0, 1$, *with entire maps* g_0, g_1, *respectively. For every* $\epsilon > 0$ *there is a smooth family* $\widehat{h}_\tau$, *holomorphic in* (y,u) *on a common neighborhood of* $B \times P$, *which equals* g_0, g_1 *on endpoint intervals and satisfies*

$$\sup_{[0,1]\times C^- \times P} d_Y(\widehat{h}_\tau, h_\tau) < \epsilon.$$

For $s = 0$, *the convex and polydisc factors in* $\mathbb{C}^s$ *are omitted.*

Proof. Suppose first that $s > 0$. Choose $d \in \operatorname{int} C$ and $L > 1$ so large that $d + L^{-1}(B - d) \Subset \operatorname{int} C$. Take smooth functions $\kappa_0, \kappa_1 : [0,1] \to [0,1]$ with disjoint supports inside the intervals where $h_\tau = g_0, g_1$, respectively, and equal to one near the corresponding endpoints. Set

$$\lambda(\tau) = 1 + (L-1)(\kappa_0(\tau) + \kappa_1(\tau)), \qquad \widetilde{h}_\tau(y,u) = h_\tau(d + \lambda(\tau)(y-d), u).$$

This family is defined near $[0,1] \times C \times P$: wherever $\lambda > 1$, the map being evaluated is an entire endpoint map. Fix chart sprays of g_0, g_1 over a neighborhood of $B \times P$ using Lemma 2.3.

Apply Lemma 6.3 to $\widetilde{h}$ and then Lemma 6.4 under A_s. This gives an entire $E(\tau, y, u)$ arbitrarily close to $\widetilde{h}$ on $[0,1] \times C \times P$. Set

$$A_\tau(y,u) = E(\tau, d + \lambda(\tau)^{-1}(y-d), u).$$

For $y \in C^-$ the rescaled argument lies in $\operatorname{conv}(C^- \cup \{d\}) \Subset \operatorname{int} C$. Substitution into $\widetilde{h}_\tau$ gives exactly $h_\tau(y,u)$. Consequently A_τ approximates h_τ uniformly on the required smaller product. On endpoint intervals where $\lambda = L$, the same identity holds for every $y \in B$, and $h_\tau = g_\ell$ there. Thus A_τ is arbitrarily close to g_ℓ on the full product $B \times P$ on those intervals.

The fixed endpoint charts therefore give small fiber coordinates $\beta_\tau^{(\ell)}$ such that

$$A_\tau = (g_\ell)^\sharp(\,\cdot\,, \beta_\tau^{(\ell)}).$$

On a smaller closed endpoint interval, choose a smooth cutoff ϑ_ℓ equal to zero near ℓ and to one near its inner end. Replace A_τ there by $(g_\ell)^\sharp(\,\cdot\,, \vartheta_\ell \beta_\tau^{(\ell)})$. The replacement agrees with A_τ near the inner end and with g_ℓ near the outer end. It is smooth in τ, holomorphic on a common neighborhood of $B \times P$, and remains arbitrarily close to h_τ on $C^- \times P$. These assertions follow from uniform continuity of the fixed charts and their inverses on compact sets. For $s = 0$, take $\lambda = 1$, omit scaling, and use the same entire approximation and endpoint repair. □

Lemma 6.6. *Assume* A_{r-1}, $r \geq 1$. *Let* $F \subset \mathbb{R} \times \mathbb{C}^{r-1}$ *be compact convex and* P *a closed polydisc. Every map holomorphic near* $F \times P$ *can be approximated uniformly there by maps entire in all displayed complex variables.*

The slices of F vary with the real coordinate t, whereas Lemma 6.3 requires a family on a fixed product. We join local entire approximants using Lemma 6.5 on a common polydisc B, then complexify the assembled family and approximate it by an entire map. Figure 4 shows one join.

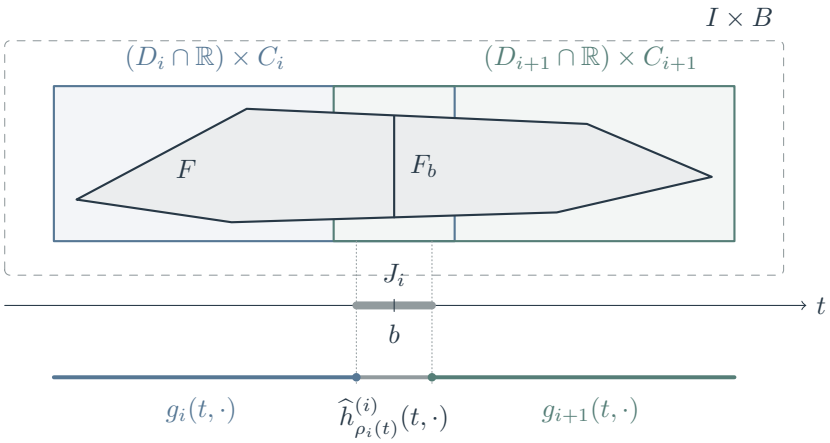


FIGURE 4. Joining local approximants along a real face. The upper panel is schematic in the y directions and omits the passive factor P. The local joining path is extended to B before it is inserted over J_i. Exact endpoint agreement makes the lower family smooth; holomorphic dependence on t is obtained afterward.

Proof. Write the map as $f(t, y, u)$. By first enlarging F slightly inside its real hyperplane, and reserving a slightly larger passive polydisc, we may suppose that its projection I to $\mathbb{R}$ has positive length. For $t \in I$, write $F_t = \{y : (t, y) \in F\}$ for the slice, and choose a compact convex neighborhood C_t of F_t and a complex disc D_t about t such that f is holomorphic near $\overline{D}_t \times C_t \times P$. Nearby slices remain in $\operatorname{int} C_t$: otherwise a convergent sequence of points in those slices would give a point of F_t outside $\operatorname{int} C_t$. A finite subdivision of I therefore assigns to each subinterval a product $D_i \times C_i \times P$ containing all its slices with positive margins.

At a division point b, shrink the joining disc and choose compact convex sets $C_b^- \Subset \operatorname{int} C_b$ inside $\operatorname{int}(C_i \cap C_{i+1})$, with every nearby slice contained in $\operatorname{int} C_b^-$. After all division points have been treated, choose a closed polydisc B containing all the C_i in its interior. Property A_{r-1}, with (t, u) passive, gives entire maps g_i as close as desired to f on the assigned products. A chart spray of f on a joining product writes the two adjacent maps as

$$g_i = f^\sharp(\,\cdot\,, a_i), \qquad g_{i+1} = f^\sharp(\,\cdot\,, a_{i+1}),$$

where both fiber coordinates are uniformly small. If $\chi : [0, 1] \to [0, 1]$ is smooth, zero near 0 and one near 1, then

$$h_\tau = f^\sharp(\,\cdot\,, (1 - \chi(\tau))a_i + \chi(\tau)a_{i+1})$$

is a local joining path uniformly close to f. Its chart and fiber coordinates are defined only on the joining product, so the path does not yet extend over B. Apply Lemma 6.5 with active variables y, with the smaller joining disc and u as passive variables, and with the common polydisc B. We obtain a path $\widehat{h}^{(i)}_\tau$ defined over B, still close to f on the slices near b, and agreeing exactly with g_i, g_{i+1} on endpoint intervals.

The path parameter τ is smooth, while t remains a passive holomorphic variable in $\widehat{h}^{(i)}_\tau$. Choose disjoint small real joining intervals J_i and smooth functions $\rho_i : J_i \to [0, 1]$ equal to 0 and 1 near their left and right ends. Define

$$H(t, y, u) = \begin{cases} \widehat{h}^{(i)}_{\rho_i(t)}(t, y, u), & t \in J_i, \\ g_i(t, y, u), & t \text{ between successive joining intervals.} \end{cases}$$

Exact endpoint agreement makes H smooth in t and holomorphic in (y, u) on a neighborhood of $I \times B \times P$. Extend it a little past the endpoints of I using the first and last entire maps. It approximates f uniformly on $F \times P$ as closely as desired. Lemma 6.3 now makes H holomorphic

near the whole product. Apply Lemma 6.4 with $s = 0$, since $B \times P$ is a polydisc, to obtain the required entire map. Here A_0 follows from A_{r-1} by taking the active convex factor to be a point and restricting the resulting entire map to that point. When $r = 1$, omit y and the corresponding convex factors throughout the construction. □

6.3. **Fixing the collar before gluing.** Kusakabe [44] uses a related construction combining a local fiberwise right inverse, Oka–Weil approximation, and gluing of sprays. A close approximation on a real face need not be close on a predetermined complex collar. The accuracy of the first entire approximation is used only to obtain a fiber inverse on some collar. After fixing this collar and its splitting constant, polynomial approximation of the fixed inverse makes the gluing error arbitrarily small.

Lemma 6.7. *Let $M \subset \mathbb{C}^m$ be compact and $F : \mathrm{Op}(M) \times \Delta^n_{4b} \to Y$, $b > 0$, be a chart spray, so $(Z, v) \mapsto (Z, F(Z, v))$ is biholomorphic onto its image. If an entire map $G : \mathbb{C}^m \times \mathbb{C}^n \to Y$ is sufficiently close to F on $M \times \overline{\Delta}^n_{3b}$, then there are a complex neighborhood U of M and a holomorphic map $W : U \times \Delta^n_{2b} \to \mathbb{C}^n$ such that*

$$G(Z, W(Z, v)) = F(Z, v). \tag{6.1}$$

For every compact convex $C \subset U$, polynomial maps $q(Z, v)$ can make the transition

$$F(Z, \cdot)^{-1} G(Z, q(Z, v)) = v + c(Z, v)$$

well-defined near $C \times \overline{\Delta}^n_b$, with both $\sup \|c\|$ and $\sup \|D_v c\|$ arbitrarily small there. The maps F, G, W, the neighborhood U, and the compact set C remain fixed during this final approximation.

Proof. For sufficiently close approximation the old chart inverse defines

$$\Gamma(Z, w) = F(Z, \cdot)^{-1} G(Z, w)$$

near $M \times \overline{\Delta}^n_{3b}$, uniformly close to w. Cauchy estimates only in w give, after choosing the approximation sufficiently accurate,

$$\|\Gamma - \mathrm{id}\| < b/4, \qquad \|D_w \Gamma - I\| < 1/2$$

on $M \times \overline{\Delta}^n_{5b/2}$, using the maximum norm and its induced operator norm. Fix G. The strict inequalities persist on $U \times \overline{\Delta}^n_{5b/2}$ for a complex neighborhood U of M. For $v \in \Delta^n_{2b}$, the map

$$w \longmapsto v - (\Gamma(Z, w) - w)$$

is a contraction of $\overline{\Delta}^n_{5b/2}$ into itself. Its unique fixed point $W(Z, v)$ satisfies (6.1). The holomorphic implicit function theorem and uniqueness give joint holomorphicity in (Z, v).

Now fix a compact convex $C \subset U$. Approximate W componentwise by polynomials on the compact convex set $C \times \overline{\Delta}^n_{3b/2}$. For these polynomial maps q, (6.1) and uniform continuity of the fixed G show that $G(Z, q(Z, v))$ converges uniformly to $F(Z, v)$ there. The old chart inverse consequently defines a transition converging uniformly to v. Cauchy estimates across the fixed fiber gap from $3b/2$ to b give uniform convergence of $D_v c$ to zero as well. The compositions $G(Z, q(Z, v))$ are entire, leaving the construction unrestricted away from this compact overlap. □

Lemma 6.8. *Assume A_{r-1}, $r \geq 1$. Let $Q \subset \mathbb{C}^r$ be compact convex and P a closed polydisc. Set $Q_- = Q \cap \{\Im z_1 \leq 0\}$ and suppose that Q_- has nonempty interior. Every map holomorphic near $Q_- \times P$ can be approximated uniformly there by maps holomorphic near $Q \times P$.*

Proof. There is nothing to prove if $Q_- = Q$. Otherwise the cutting hyperplane meets $\operatorname{int} Q$. Set $K = Q \times P$ and $Z = (z, u)$. By Lemma 2.3, fix a chart spray $F(Z, v)$ centered at the given map f over a neighborhood of $K_- = Q_- \times P$, with fiber polydisc Δ^n_{4b}. Choose an enlarged compact convex real face $\Sigma^+ \subset \mathbb{R} \times \mathbb{C}^{r-1}$ and an enlarged passive polydisc P^+ such that

$$M = \Sigma^+ \times P^+$$

contains $K \cap \{\Im z_1 = 0\}$ in its relative interior and F is holomorphic near $M \times \overline{\Delta}^n_{3b}$. Lemma 6.6, with (u, v) passive, supplies an entire $G(Z, w)$ sufficiently close to F on that product for Lemma 6.7. Fix this G and the resulting inverse W on $U \times \Delta^n_{2b}$.

Choose a bounded smoothly bounded strongly convex domain $D \supset K$ sufficiently close to K that its closed lower part lies in the base domain of F and its closed section by $\Im z_1 = 0$ lies in U. Compactness then gives $\delta > 0$ such that

$$\overline{D} \cap \{\Im z_1 \le 2\delta\} \subset \mathrm{dom}_Z\, F, \qquad C = \overline{D} \cap \{|\Im z_1| \le 2\delta\} \Subset U.$$

Set

$$A = D \cap \{\Im z_1 < \delta\}, \qquad B = D \cap \{\Im z_1 > -\delta\}.$$

These domains have separated closed fringes, $A \cup B = D$, and $\overline{A \cap B} \subset C$, where C is compact convex. In particular, they satisfy the bounded center-gluing hypotheses of Lemma 2.5. Fix the domains, their splitting constant, and all fiber margins.

Figure 5 summarizes the source geometry and the order of these choices; the other source coordinates are suppressed in the diagram.

Lemma 6.7 now gives a polynomial q for which the sprays

$$F_A(Z, v) = F(Z, v), \qquad F_B(Z, v) = G(Z, q(Z, v))$$

have transition $v + c(Z, v)$ on the overlap, with both c and $D_v c$ as small as desired on the fixed fiber polydisc Δ^n_b. The second family is entire. Lemma 2.5 gives small holomorphic sections a on A and b_1 on B satisfying

$$F_A(Z, a(Z)) = F_B(Z, b_1(Z)) \quad (Z \in A \cap B).$$

Their centers define a holomorphic map on D. As the polynomial approximation improves, $\|a\|_A \to 0$, and hence this map converges uniformly to $f = F(\,\cdot\,, 0)$ on $K_- \Subset A$. Since D contains K, the assertion follows. □

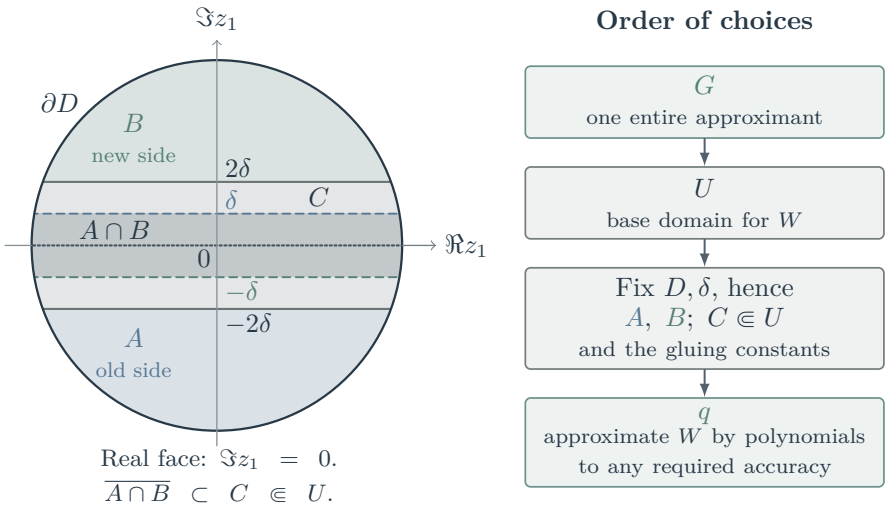


FIGURE 5. A schematic slice of the one-cut construction. The dashed lines bound the open overlap $A \cap B$; the solid lines bound the larger compact collar C, which includes the intervening boundary points of D. The central dotted line is the real face. The entire approximant is fixed before the collar geometry, and only then is the inverse approximation refined.

6.4. **Finite removal of the cuts.** A polytope inside the original domain provides the initial compact set; successively dropping its defining inequalities enlarges it to a polydisc.

Proof of Theorem 6.2. Assume inductively that A_{r-1} of Definition 6.1 holds. Let $K \subset \mathbb{C}^r$ be compact convex, P a closed polydisc, and f holomorphic near $K \times P$. Fix $\epsilon > 0$ and choose a full-dimensional compact real convex polytope K' such that

$$K \Subset \operatorname{int} K', \qquad K' \times P \Subset \operatorname{dom} f.$$

Write $K' = \bigcap_{j=1}^{N} \{\ell_j \le c_j\}$ with real affine functionals ℓ_j, and choose R with $K' \Subset \operatorname{int} \overline{\Delta}_R^r$. Define

$$Q_k = \overline{\Delta}_R^r \cap \bigcap_{j=k+1}^{N} \{\ell_j \le c_j\}, \qquad 0 \le k \le N.$$

Then $Q_0 = K'$, $Q_N = \overline{\Delta}_R^r$, and $Q_{k-1} = Q_k \cap \{\ell_k \le c_k\}$. Each Q_{k-1} has nonempty interior because it contains K'.

For a nonconstant $\ell_k(z) = \Re \sum_j a_j z_j + d$, the affine coordinate $w_1 = i(\sum_j a_j z_j + d - c_k)$ transforms its inequality into $\Im w_1 \le 0$; complete the linear part to a complex coordinate system. Convexity is preserved and the passive polydisc is unchanged. Thus Lemma 6.8 applies at each nontrivial step; a redundant cut needs no change of map. Starting from $f_0 = f$, obtain maps f_k holomorphic near $Q_k \times P$ such that

$$\sup_{Q_{k-1} \times P} d_Y(f_k, f_{k-1}) < \epsilon_k, \qquad \sum_{k=1}^{N} \epsilon_k < \epsilon/2.$$

Finally, A_0 approximates f_N on the polydisc $\overline{\Delta}_R^r \times P$ by an entire map within $\epsilon/2$. The triangle inequality gives approximation to f within ϵ on $K \times P$. This proves A_r and completes the induction. Taking the passive factor to be a point gives the convex approximation property. □

Completion of the proofs of Theorems 1.3 and 1.4. LAP gives local planar extension by Lemma 3.4, polydisc approximation by Theorem 5.4, and convex approximation by Theorem 6.2. This proves the LAP criterion. Proposition 2.2 therefore makes every strongly dominable manifold Oka. The converse is classical [25]; Proposition 2.2 then also gives LAP for every Oka manifold. In dimension zero all three properties are immediate. □

6.5. **Direct consequences.**

Proof of Corollary 1.5. Only sufficiency needs proof. For $s > 0$, let B_s denote the Euclidean ball of radius s in $\mathbb{C}^n$. Fix $x \in X$, choose $r > 0$, and take a coordinate parametrization $\phi : B_{2r} \to X$ with $\phi(0) = x$. By CAP_n, an entire map $G : \mathbb{C}^n \to X$ approximates ϕ arbitrarily closely on $\overline{B}_{3r/2}$. Choose the approximation so close that its image on this ball lies in $\phi(B_{2r})$ and, for $g = \phi^{-1} \circ G$,

$$\sup_{\overline{B}_r} |g(z) - z| < r/2, \qquad \sup_{\overline{B}_r} \|dg_z - I\| < 1/2.$$

The second estimate follows from Cauchy estimates on the larger ball. The map $z \mapsto z - g(z)$ is therefore a contraction of $\overline{B}_r$ into itself. Its fixed point a satisfies $g(a) = 0$, and dg_a is invertible. Thus $z \mapsto G(z + a)$ dominates X at x. Apply Theorem 1.4.

The same argument applies to the special convex sets in Forstnerič's original definition [23, Definition 1.1 and p. 692]: take two nested balls in the interior of such a set and rescale the coordinate parametrization to be defined near the whole set. □

Proof of Corollary 1.6. Combine the strong dominability theorem of Forstnerič and Lárusson [29] with Theorem 1.4, componentwise. □

Proof of Corollary 1.7. By Theorem 1.4, it suffices to observe that a dominating map into an open subset also dominates the ambient manifold. □

7. Removability and holomorphic images

7.1. **Removability across thin exceptional sets.** The LAP criterion also applies when LAP is known only away from a sufficiently thin set, more precisely, a set of zero $(2n-1)$-dimensional Hausdorff measure. The exponent $2n-1$ comes from the parameters of a coordinate circle about a fixed point: its complex line contributes $2n-2$ real dimensions and its radius contributes one. A set of zero $(2n-1)$-dimensional Hausdorff measure cannot meet every such circle.

For a closed exceptional set with this Hausdorff-measure bound, Alarcón and Forstnerič proved that dominability off the set implies the Oka-1 property [2]. Theorem 1.8 gives the full Oka property and also allows a nonclosed exceptional set.

We first recenter the families supplied by Lemma 5.2.

Lemma 7.1. *For $r > 0$ and closed polydiscs $P_j \subset \mathbb{C}^{n-1}$ with $0 \in \operatorname{Int} P_j$, $1 \le j \le n$, suppose that ψ_j is a biholomorphic coordinate parametrization near $\overline{\Delta}_r \times P_j$, $\psi_j(0,0) = x$, and the vectors $\partial_z \psi_j(0,0)$ form a basis of $T_x Y$. Also suppose that for some $S_\nu \to \infty$ there are holomorphic maps*

$$\sigma_{j,\nu} : \operatorname{Op}(\overline{\Delta}_{S_\nu} \times P_j) \longrightarrow Y$$

converging uniformly to ψ_j on $\overline{\Delta}_r \times P_j$. Then $x \in G_Y$.

Proof. Fix small source polydiscs Q_j around $(0,0)$, with closures in $\Delta_r \times \operatorname{Int} P_j$. On slightly larger fixed polydiscs, uniform convergence places the images of $\sigma_{j,\nu}$ in the ψ_j-chart for all sufficiently large ν. Cauchy estimates give

$$\psi_j^{-1} \circ \sigma_{j,\nu} = \mathrm{id} + e_{j,\nu}, \qquad e_{j,\nu} \longrightarrow 0 \quad \text{in } C^1(\overline{Q}_j).$$

For example, if $Q_j = \Delta^n_{\rho_j}$ after separate coordinate rescalings, arrange $\|e_{j,\nu}\| < \rho_j/4$ and $\|De_{j,\nu}\| < 1/2$ there. For a point in $\Delta^n_{\rho_j/2}$, the equation $z + e_{j,\nu}(z) = w$ has a unique solution by contraction on the closed $3\rho_j/4$ polydisc. The solutions depend holomorphically on w and converge uniformly on smaller products to w.

Intersect the resulting finitely many target neighborhoods and shrink once to a relatively compact neighborhood $V \ni x$. After discarding a common finite initial segment, this gives holomorphic branches

$$\ell_{j,\nu} = (s_{j,\nu}, p_{j,\nu}) : V \longrightarrow \Delta_r \times \operatorname{Int} P_j, \qquad \sigma_{j,\nu} \circ \ell_{j,\nu} = \mathrm{id}_V,$$

converging to ψ_j^{-1}, with images in fixed compact subproducts. In particular, $|s_{j,\nu}(y)| < C$ for one constant C, independent of j, ν, y.

Set $R_\nu = S_\nu - C$ and discard further initial terms so that $R_\nu > 0$. Define

$$h_{j,\nu}(y,t) = \sigma_{j,\nu}(s_{j,\nu}(y) + t, p_{j,\nu}(y)).$$

These maps are holomorphic on the common products $V \times \Delta_{R_\nu}$ and satisfy $h_{j,\nu}(y,0) = y$ exactly. The fixed compact margins permit one $\tau > 0$ such that, for $y \in V$ and $|t| < \tau$, all translated arguments remain inside the original convergence products. Thus the new families converge locally uniformly there to

$$h_j(y,t) = \psi_j(\psi_j^{-1}(y) + (t,0)).$$

Their initial velocities at x are $\partial_z \psi_j(0,0)$. These are precisely the data of Definition 1.2. □

Proposition 7.2. *Let ϑ be a target chart about $x \in Y$. If the closed coordinate disc $\vartheta(x) + \overline{\Delta}_r v$, $v \neq 0$, lies in the chart image and its boundary lies in $\vartheta(G_Y)$, then $x \in G_Y$.*

Proof. Compactness of the boundary and openness of G_Y permit sufficiently small perturbations of v while retaining both chart containment and the good-boundary condition. We

choose n linearly independent vectors v_j among these perturbations, complete each to a basis $v_j, e_{j,2}, \dots, e_{j,n}$, and set

$$\psi_j(z,p) = \vartheta^{-1}\left(\vartheta(x) + zv_j + \sum_{k=2}^{n} p_{k-1}e_{j,k}\right).$$

For sufficiently small transverse polydiscs P_j, these are coordinate parametrizations near $\overline{\Delta}_r \times P_j$. To choose P_j, first cover the central boundary circle by finitely many arcs whose images lie relatively compactly in neighborhoods supplied by Lemma 3.4, and then shrink the transverse polydisc so that every corresponding closed arc product lies in its chosen neighborhood. This choice is independent of all later approximation indices.

For any common sequence $S_\nu \to \infty$, Lemma 5.2 gives maps $\sigma_{j,\nu}$ holomorphic near $\overline{\Delta}_{S_\nu} \times P_j$, converging uniformly to ψ_j on the original closed product. Their limiting active velocities are $(d\vartheta_x)^{-1}v_j$, a basis. Lemma 7.1 now gives $x \in G_Y$. □

Proof of Theorem 1.8. By hypothesis, $Y \setminus E \subset G_Y$. Fix $x \in Y$ and take coordinates centered at x. Choose $0 < a < b$ so that the closed ball of radius b lies in the chart. On the annulus $a \le |z| \le b$, the smooth map

$$q(z) = ([z], |z|) \in \mathbb{P}^{n-1} \times [a,b]$$

is Lipschitz, so the image of the coordinate representative of E there has zero $(2n-1)$-dimensional Hausdorff outer measure. The manifold $\mathbb{P}^{n-1} \times (a,b)$ has positive measure in that dimension. Choose $([v], r)$ outside this image, with $|v| = 1$. Its fiber is the circle $\{\zeta v : |\zeta| = r\}$, whose inverse image under the chart lies in G_Y; the filled disc lies in the chart image. Proposition 7.2 gives $x \in G_Y$. This also covers $n = 1$, when $\mathbb{P}^0$ is a point. Thus $G_Y = Y$, and Theorem 1.3 makes Y Oka. □

Proof of Theorem 1.9. Set $n = \dim Y$. The case $n = 0$ is immediate. For $n \ge 1$, every subset of a proper complex analytic subset of a connected n-manifold has zero $(2n-1)$-dimensional Hausdorff outer measure. A countable cover by the given neighborhoods therefore shows that $Y \setminus G_Y$ has zero measure in this dimension, so Theorem 1.8 applies. For the particular case, dominating maps into $X \setminus A$ also dominate X after composition with the inclusion. Thus Proposition 2.2 gives LAP on $X \setminus A$, and the first assertion applies. □

By Proposition 2.2, Theorem 1.8 also applies when Y is dominable at every point outside E. In particular, if such an exceptional set E is closed and $Y \setminus E$ is Oka, then Y is Oka.

The following result was proved by Wang, Xie, and Zhao [55, Corollary 1.3]; it also follows from OpenAI's theorem that every complex K3 surface is Oka [49, Theorem 1.1].

Corollary 7.3. *Every complex Kummer surface is Oka.*

Proof. For a Kummer surface X, the complement of its sixteen exceptional curves is Oka by Forstnerič and Lárusson [29, Lemma 7], so Theorem 1.9 applies. □

For every closed subset K of an affine complex hyperplane $H \subset \mathbb{C}^n$, $n \ge 1$, the complement $\mathbb{C}^n \setminus K$ is Oka. Indeed, apply Theorem 1.8 with $Y = \mathbb{C}^n \setminus K$ and $E = H \setminus K$. The set E has zero $(2n-1)$-dimensional Hausdorff measure, and $Y \setminus E = \mathbb{C}^n \setminus H \cong \mathbb{C}^{n-1} \times \mathbb{C}^*$ is Oka. Set $T = \{0\} \cup \{1/j : j \ge 1\}$. For $n \ge 3$, this includes

$$\mathbb{C}^n \setminus (T^2 \times \{0\}^{n-2}).$$

A complex manifold Y is *weakly subelliptic* if, for every compact set $K \subset Y$, there are finitely many holomorphic sprays on Y whose vertical differentials jointly span T_yY at every $y \in K$. Kusakabe proved that this example is Oka but not weakly subelliptic [46].

Remark 7.4. A complex manifold has the *density property* if every holomorphic vector field can be approximated uniformly on compact sets by Lie combinations of holomorphic vector fields whose flows exist for every complex time. Another direct recovery is Kusakabe's theorem on compact holomorphically convex complements [45]. If X is a Stein manifold of dimension $n > 1$ with the density property and $K \subset X$ is compact and $\mathcal{O}(X)$-convex, the construction of Forstnerič and Wold places every point of $X \setminus K$ in an open subset biholomorphic to $\mathbb{C}^n$ [35]. Thus $X \setminus K$ is strongly dominable, and Theorem 1.4 makes it Oka.

7.2. **Surjective holomorphic images, complements, and group orbits.** The proof for a surjective holomorphic map into Y uses the fact that its critical values have zero $(2\dim Y - 1)$-dimensional Hausdorff measure, even when the map is not proper.

7.2.1. *Surjective holomorphic images.*

Theorem 7.5. *Let $p : X \to Y$ be a surjective holomorphic map of connected complex manifolds. If X is Oka, then Y is Oka.*

Proof. Set $m = \dim X$ and $n = \dim Y$. The case $n = 0$ is immediate. For $n \geq 1$, let

$$C_p = \{x \in X : \operatorname{rank} dp_x < n\}, \qquad E = p(C_p).$$

The closed analytic set C_p admits a locally finite stratification by complex submanifolds on which the restrictions of p have constant rank [52, Chapter 2, Proposition 3]. There are countably many strata, and each has a countable constant-rank coordinate cover. On every such coordinate neighborhood the image is contained in a complex submanifold of Y of dimension at most $n - 1$. Hence $\mathcal{H}^{2n-1}(E) = 0$ in the outer-measure sense.

For $y \in Y \setminus E$, choose $x \in p^{-1}(y)$. Strong dominability of X gives an entire map $F : \mathbb{C}^m \to X$ with $F(0) = x$ and dF_0 invertible. Since dp_x is surjective, there is a linear map $\iota : \mathbb{C}^n \to \mathbb{C}^m$ for which $d(p \circ F)_0 \circ \iota$ is an isomorphism. Thus $p \circ F \circ \iota$ dominates Y at y. Theorem 1.8 completes the proof. □

Corollary 7.6. *Let $\pi : \widetilde{X} \to X$ be a proper bimeromorphic holomorphic map of connected complex manifolds. If $\widetilde{X}$ is Oka, then X is Oka. In particular, this holds when $\widetilde{X} = \mathrm{Bl}_Z X$ for a smooth closed complex submanifold $Z \subsetneq X$.*

Proof. This follows directly from Theorem 7.5, since a proper bimeromorphic map is surjective. □

This answers the blow-down question of Forstnerič and Lárusson [29, Section 6].

Following Forstnerič [32, Definition 1.2], a connected complex manifold Y has the *basic Oka property with surjectivity* (BOPS) if every continuous map $S \to Y$ from a Stein manifold with $\dim S \geq \dim Y$ is homotopic to a surjective holomorphic map.

Corollary 7.7. *For a connected complex manifold Y of dimension $n \geq 1$, the following conditions are equivalent:*

(i) *Y is Oka;*
(ii) *there is a surjective holomorphic map $\mathbb{C}^n \to Y$;*
(iii) *there is a surjective holomorphic map $\mathbb{C}^N \to Y$ for some finite integer $N \geq 1$;*
(iv) *Y has BOPS.*

Proof. Forstnerič's surjectivity theorem gives both (ii) and (iv) from (i) [32, Theorem 1.1]. Clearly (ii) implies (iii), while Theorem 7.5 gives (iii)⇒(i). Applying BOPS to a constant map $\mathbb{C}^n \to Y$ gives (iv)⇒(ii). □

This answers Forstnerič's question whether BOPS implies the Oka property [32, Problem 1.4(a)].

7.2.2. *Product complements and blowups.*

Theorem 7.8. *Let X and Y be connected positive-dimensional Oka manifolds, with $A \subsetneq X$ and $B \subsetneq Y$ closed complex analytic subsets. If $Z \subset X \times Y$ is a closed complex analytic subset contained in $A \times B$, then $(X \times Y) \setminus Z$ is Oka.*

Proof. First consider $Z = A \times B$ and fix (p, q) in its complement. After exchanging the factors, assume $p \notin A$. Set $n = \dim X$ and $m = \dim Y$. Choose entire maps $f : \mathbb{C}^n \to X$ and $g : \mathbb{C}^m \to Y$, centered at p and q and nonsingular at zero. The image of g contains a neighborhood of q. Since B has empty interior, some $c \in \mathbb{C}^m$ satisfies $g(c) \notin B$.

The analytic set $f^{-1}(A)$ avoids zero. Cartan's theorem A applied to its ideal sheaf gives $h \in \mathcal{O}(\mathbb{C}^n)$ with $h|_{f^{-1}(A)} = 0$ and $h(0) = 1$ [25]. Define

$$\Phi(u, v) = (f(u), g(c + h(u)(v - c))).$$

Whenever $f(u) \in A$, the second component equals $g(c) \notin B$. Thus Φ takes values in the complement and $\Phi(0, 0) = (p, q)$. Its differential at zero is block triangular with invertible diagonal blocks df_0 and dg_0. The complement is strongly dominable, hence Oka.

For general Z, the manifold $W = (X \times Y) \setminus Z$ contains the connected dense Oka open set $U = (X \times Y) \setminus (A \times B)$. The complement $W \setminus U$ is proper and closed analytic in the connected manifold W. Theorem 1.9 therefore makes W Oka. □

Corollary 7.9. *Let X, Y, A, B be as in Theorem 7.8.*

(i) *If $Z \subset A \times B$ is a smooth closed complex submanifold of $X \times Y$, possibly disconnected, then $\mathrm{Bl}_Z(X \times Y)$ is Oka. In particular, this applies to $Z = A \times B$ when both A and B are smooth.*

(ii) *For every finite set $D \subset X \times Y$, the complement $(X \times Y) \setminus D$ and the blowup of $X \times Y$ at the distinct points of D are Oka.*

Proof. For a nonempty smooth center Z, the blowup is a connected complex manifold whose exceptional divisor has complement biholomorphic to $(X \times Y) \setminus Z$. The latter is Oka by Theorem 7.8, so Theorem 1.9 applies upstairs. An empty center gives the Oka product.

For a nonempty finite set D, its projections $A_D = \mathrm{pr}_X(D)$ and $B_D = \mathrm{pr}_Y(D)$ are proper closed analytic subsets because both factors have positive dimension. Since $D \subset A_D \times B_D$, the theorem gives the Oka complement, and (i) gives the Oka blowup. The case $D = \varnothing$ is immediate. □

7.2.3. *Open group orbits.*

Corollary 7.10. *Let a complex Lie group G act holomorphically on a connected complex manifold X. If one G-orbit is open, then X is Oka.*

Proof. The zero-dimensional case is immediate. For $n = \dim X > 0$, take a basis $\xi_1, \dots, \xi_N$ of the Lie algebra of G, and let $V_1, \dots, V_N$ be the induced holomorphic vector fields. The set

$$A = \{x \in X : \mathrm{rank}(V_1(x), \dots, V_N(x)) < n\}$$

is closed analytic, since it is defined by the n-fold exterior products of these fields. It is proper because the fields span the tangent spaces along the open orbit.

At any $x \notin A$, choose n of the fields forming a basis of $T_x X$, with indices $j_1, \dots, j_n$. The entire map

$$(t_1, \dots, t_n) \longmapsto \exp(t_n \xi_{j_n}) \cdots \exp(t_1 \xi_{j_1}) \cdot x$$

dominates X at x. A proper analytic subset of the connected n-manifold X has zero $(2n - 1)$-dimensional Hausdorff measure. Theorem 1.8 now applies. □

8. Rationally connected projective manifolds

Campana and Winkelmann constructed dense entire curves in rationally connected projective manifolds whose images contain any prescribed countable subset [8]. Alarcón and Forstnerič conjectured that these manifolds are Oka-1 [2]; Du, Guo, and Xie proved this in a recent preprint [17]. Theorem 1.10 gives the full Oka property, settling the rationally connected case of Forstnerič's question on Oka projective manifolds [37].

For a complex manifold X of dimension n, a rational curve $g : \mathbb{P}^1 \to X$ is *very free* if g^*TX is ample. By the splitting theorem on $\mathbb{P}^1$, this means

$$g^*TX \simeq \bigoplus_{i=1}^{n} \mathcal{O}_{\mathbb{P}^1}(a_i), \qquad a_i \geq 1.$$

Very free rational curves provide the centered entire families required by the analytic criterion. The essential deformation statement concerns first jets: one local section of first-jet evaluation supplies all tangent directions at once.

Fix distinct points $0, \infty \in \mathbb{P}^1$ and an affine coordinate t on $\mathbb{P}^1 \setminus \{\infty\}$ with $t(0) = 0$. For a parametrized curve g we write $g'(0) = dg_0(\partial/\partial t)$. The space of first jets of maps from this fixed pointed source to a manifold Y is TY: a jet records its value and its velocity.

Lemma 8.1. *Let Y be a complex n-manifold, S a complex manifold, and $F : S \times \mathbb{C} \to Y$ a holomorphic family. Suppose*

$$J : S \longrightarrow TY, \qquad J(s) = \big(F(s,0), \partial_t F(s,0)\big)$$

is a submersion at some $s_0 \in S$. Then, on a neighborhood V of $x = F(s_0, 0)$, there are holomorphic maps $h_j : V \times \mathbb{C} \to Y$, $1 \leq j \leq n$, such that $h_j(y, 0) = y$ and the vectors $\partial_t h_j(y, 0)$ form a frame of $TY|_V$. If F extends to $S \times \mathbb{P}^1$, so do these families.

Proof. Choose a local holomorphic section $\sigma : W \to S$ of J, where $W \subset TY$ is a neighborhood of $J(s_0)$. Every nonempty open subset of a finite-dimensional vector space contains a basis, so choose $v_1, \ldots, v_n \in W \cap T_x Y$ forming a basis. Extend them to a local holomorphic frame $v_j(y)$ and shrink V until all its graphs lie in W. Set

$$h_j(y, t) = F(\sigma(y, v_j(y)), t).$$

The identity $J \circ \sigma = \mathrm{id}_W$ gives the asserted values and velocities. If F extends to $S \times \mathbb{P}^1$, the same formula defines h_j on $V \times \mathbb{P}^1$. □

We use the strong form of rational connectedness: a smooth projective rationally connected variety over $\mathbb{C}$ admits a very free curve through every prescribed point [19, Theorem 2.40].

Lemma 8.2. *Let $g : \mathbb{P}^1 \to X$ be very free, where X is a smooth projective complex manifold. In the scheme H of morphisms of its fixed projective degree, both H and the first-jet morphism*

$$J : H \longrightarrow TX, \qquad u \longmapsto (u(0), u'(0)),$$

are smooth at g.

Proof. Let $2[0]$ denote the length-two subscheme defined by $t^2 = 0$, and set $E = g^*TX$. The splitting above gives $H^1(\mathbb{P}^1, E) = H^1(\mathbb{P}^1, E(-2[0])) = 0$. Fixed-source deformation theory identifies $T_g H$ with $H^0(\mathbb{P}^1, E)$ and makes H smooth at g [19, Proposition 1.6 and Theorem 1.9]. Infinitesimal deformation of a first jet identifies $T_{J(g)}TX$ with $H^0(2[0], E|_{2[0]})$, and dJ_g is restriction. The sequence

$$0 \longrightarrow E(-2[0]) \longrightarrow E \longrightarrow E|_{2[0]} \longrightarrow 0$$

therefore makes dJ_g surjective. Since source and target are smooth at the relevant points, J is smooth at g. □

Proof of Theorem 1.10. The zero-dimensional case is immediate. Otherwise fix $x \in X$ and choose one very free curve g with $g(0) = x$. Lemma 8.2 and universal evaluation on $H \times \mathbb{P}^1$, restricted to a smooth neighborhood of g, meet the hypotheses of Lemma 8.1. We obtain centered entire families on a common neighborhood of x whose initial velocities span T_xX. Since x was arbitrary, Lemma 2.1 and Theorem 1.3 apply. The Fano assertion follows from the rational connectedness theorem of Campana and Kollár–Miyaoka–Mori [11, 42]. □

The same argument survives deletion of an algebraic subset of codimension at least two. To obtain families at every remaining point, we first find an avoiding very free curve through that point.

Lemma 8.3. *Let X be as in Theorem 1.10 and $A \subset X$ a closed algebraic subset whose irreducible components have codimension at least two. For every $x \in X \setminus A$ there is a very free curve $g : \mathbb{P}^1 \to X$ with $g(0) = x$ and $g(\mathbb{P}^1) \cap A = \varnothing$.*

Proof. Choose a very free curve through x. Very freeness is open in its fixed-degree morphism scheme, by semicontinuity of $H^1(\mathbb{P}^1, g^*TX(-2[0]))$. Evaluation at 0 is smooth there, with fiber tangent space $H^0(\mathbb{P}^1, g^*TX(-[0]))$ [19, Section 1.5, paragraph 1.10]. Take a nonempty smooth irreducible open subset S of this fiber within the very-free locus. For every $g \in S$ and $t \neq 0$, the vanishing

$$H^1\big(\mathbb{P}^1, g^*TX(-[0] - [t])\big) = 0$$

shows that evaluation of these tangent sections at t is surjective. Thus universal evaluation $e : S \times (\mathbb{P}^1 \setminus \{0\}) \to X$ is smooth.

The incidence set $I = \{(g, t) : g(t) \in A\}$ is closed in the full product $S \times \mathbb{P}^1$ and misses $S \times \{0\}$. Smoothness on its containing open set gives

$$\dim I \leq \dim S + 1 - 2 = \dim S - 1.$$

The projection $S \times \mathbb{P}^1 \to S$ is proper, so its image of I is a proper closed subset. Any parameter outside this image gives the required curve. □

Corollary 8.4. *Under the hypotheses of Lemma 8.3, $X \setminus A$ is Oka.*

Proof. Fix $x \in X \setminus A$ and choose an avoiding very free curve through x. Avoidance of A on all of $\mathbb{P}^1$ is an open condition in the unrestricted morphism scheme: its complement is the proper projection of the closed universal incidence set. On this open locus, Lemma 8.2 still makes first-jet evaluation a submersion near the chosen curve. Lemma 8.1 now gives families into $X \setminus A$. They exist at every point, so Lemma 2.1 and Theorem 1.3 prove the claim. □

8.1. **Ellipticity and birational models.**

Corollary 8.5. *Every smooth connected complex projective rationally connected manifold, in particular every complex projective Fano manifold, is holomorphically elliptic. Every smooth projective birational model of such a manifold is also Oka and holomorphically elliptic.*

Proof. Theorem 1.10 gives the Oka property. The theorem of Forstnerič–Lárusson that every projective Oka manifold is holomorphically elliptic [31] gives the first assertion. Rational connectedness is invariant under birational maps between smooth projective manifolds [19, Remark 2.21], so the same argument applies to every indicated birational model. □

8.2. **Compact Kähler consequences.**

Two structure theorems reduce the following compact Kähler cases to the projective rationally connected case.

Corollary 8.6. *Every compact Kähler manifold admitting a Kähler metric with nonnegative holomorphic sectional curvature is Oka.*

Proof. Let X be such a manifold; we may assume that it is connected. Matsumura's structure theorem [47, Theorem 1.1 and Remark 1.2] supplies an integer $q \geq 0$ and a smooth projective rationally connected manifold F, possibly a point, such that the universal cover of X is biholomorphic to $\mathbb{C}^q \times F$. By Theorem 1.10, F is Oka. Products of Oka manifolds are Oka, and the Oka property descends through unramified holomorphic coverings [24, Corollary 1.3]. Hence X is Oka. □

On a compact complex manifold Z with a fixed Hermitian form ω, a holomorphic line bundle L is *nef* if, for every $\epsilon > 0$, it admits a smooth Hermitian metric whose normalized real Chern curvature is bounded below by $-\epsilon\omega$. For a holomorphic vector bundle $E \to Z$, let $\mathbb{P}(E)$ parametrize its one-dimensional quotients. The bundle E is *nef* if $\mathcal{O}_{\mathbb{P}(E)}(1)$ is nef [18, Definitions 1.2 and 1.9].

Corollary 8.7. *Every compact Kähler manifold with nef tangent bundle is Oka.*

Proof. Let X be such a manifold; we may assume that it is connected. The structure theorem of Demailly–Peternell–Schneider [18, Theorem 3.14 and Proposition 3.9] gives a connected finite étale cover $p : X' \to X$ whose Albanese map

$$a : X' \longrightarrow \mathrm{Alb}(X')$$

is a surjective holomorphic submersion with connected Fano fibers. If the fibers are points, a is an isomorphism, so X' is a complex torus and is Oka.

Otherwise $TX' = p^*TX$ is nef, and hence so is $-K_{X'} = \det TX'$. The general fiber of a is Fano, so its anticanonical bundle is nef and big. Cao–Höring's local-triviality theorem [12] therefore makes a a holomorphic fiber bundle. Its fibers are Oka by Theorem 1.10, and its base is a complex torus. The Oka fiber bundle theorem [24, Corollary 1.3] shows that X' is Oka. In either case the Oka property descends along the unramified covering p. □

9. Divisor complements

9.1. **Logarithmic rational curves and divisor complements.** Let $D = \sum_{i=1}^{r} D_i$ be a reduced simple normal crossings divisor on a smooth projective manifold X of dimension n. Thus every D_i is smooth and locally their union is a union of coordinate hyperplanes. The bundle $T_X(-\log D)$ consists of vector fields tangent to all D_i; in coordinates where $D = \{x_1 \cdots x_k = 0\}$ it has frame

$$x_1\frac{\partial}{\partial x_1}, \ldots, x_k\frac{\partial}{\partial x_k}, \frac{\partial}{\partial x_{k+1}}, \ldots, \frac{\partial}{\partial x_n}.$$

We allow $D = 0$ and use the fixed source coordinate of Section 8.

For a divisor complement, deformation of an ordinary rational curve need not preserve avoidance of the boundary. The appropriate parameter space keeps the complete pullback divisor fixed at infinity. Its infinitesimal kernel is the logarithmic tangent bundle, and positivity of that bundle again gives a submersive first-jet map. Here the hypothesis supplies just one curve. Its deformations give families on a nonempty Zariski-open subset of the complement, after which analytic propagation reaches every point. We use the standard logarithmic deformation principle recalled in [14, Section 2.2]; here the contact conditions are realized as a fiber of an ordinary divisor map, and first-jet smoothness is checked directly.

Theorem 9.1. *Let X be a connected complex projective manifold and D a reduced simple normal crossings divisor. Suppose there is a nonconstant map $f : \mathbb{P}^1 \to X$ such that*

$$f^{-1}(D) \subseteq \{\infty\}, \qquad f^*T_X(-\log D) \text{ is ample.}$$

Then $U = X \setminus D$ is Oka.

The proof follows Lemma 9.2 below.

9.1.1. *Fixed contact and first jets.* Write $f^*D_i = m_i[\infty]$, where $m_i \geq 0$.

Fix a projective embedding of X and work in the finite-type scheme of morphisms $\mathbb{P}^1 \to X$ of degree $\deg f^*\mathcal{O}_X(1)$ [19]. Retain its open-and-closed locus where $\deg g^*\mathcal{O}_X(D_i) = m_i$ for each i, and let H be the open subset of maps contained in no D_i. For $m \geq 0$, the symmetric power $\mathrm{Sym}^m(\mathbb{P}^1)$ parametrizes effective divisors of degree m; the factor for $m = 0$, and an empty product, is a point. There is an algebraic divisor map

$$c : H \longrightarrow B := \prod_{i=1}^{r} \mathrm{Sym}^{m_i}(\mathbb{P}^1), \qquad c(g) = (g^*D_i)_i.$$

To obtain c scheme-theoretically, pull back the defining section of each D_i along universal evaluation. It is not identically zero on any source fiber. Its zero scheme is therefore a relative effective Cartier divisor, flat over H, by the fiberwise Cartier criterion [54]. It is proper and fiberwise finite of constant degree m_i, so it determines the indicated morphism to the symmetric power of the smooth curve $\mathbb{P}^1$ [41, Remark 9.3.9].

Set $b_0 = (m_i[\infty])_i$ and take the scheme-theoretic fiber

$$M = c^{-1}(b_0).$$

Thus all contact points, including their multiplicities, are fixed, while the maps remain parametrized. Universal evaluation restricts to

$$\mathcal{F} : M \times \mathbb{A}^1 \longrightarrow U. \tag{9.1}$$

The fixed coordinate identifies first jets at 0 with TU, giving an algebraic morphism $J : M \to TU$, $J(g) = (g(0), g'(0))$.

Lemma 9.2. *If $g \in M$ and $E_g = g^*T_X(-\log D)$ is ample, then $J : M \to TU$ is smooth at g. Moreover M is smooth at g and $T_gM = H^0(\mathbb{P}^1, E_g)$.*

Proof. We first prove smoothness of the contact fiber and compute its tangent space. Set

$$Z_i = m_i[\infty], \qquad Q_g = \bigoplus_{i:m_i>0} g^*\mathcal{O}_X(D_i)|_{Z_i}.$$

The logarithmic inclusion gives an exact sequence

$$0 \longrightarrow E_g \longrightarrow g^*TX \longrightarrow Q_g \longrightarrow 0. \tag{9.2}$$

Indeed, near $g(\infty)$ take normal crossings coordinates and a source coordinate z vanishing at infinity. For the components met by g, one has $x_i \circ g = z^{m_i}u_i(z)$ with $u_i(0) \neq 0$. In the corresponding normal direction, the inclusion is multiplication by $z^{m_i}u_i(z)$, with quotient modulo z^{m_i}. All other directions are unchanged. Away from infinity the inclusion is an isomorphism. This proves both injectivity and the asserted torsion quotient.

The tangent space to the divisor target is

$$T_{Z_i}\,\mathrm{Sym}^{m_i}(\mathbb{P}^1) = H^0(Z_i, \mathcal{O}_{\mathbb{P}^1}(Z_i)|_{Z_i}) = H^0(Z_i, g^*\mathcal{O}_X(D_i)|_{Z_i}).$$

The final identification uses the defining section with zero divisor Z_i. With the corresponding convention for infinitesimal divisor displacement, dc_g is the quotient map to $H^0(Q_g)$: a variation of $x_i \circ g$ is taken modulo z^{m_i}. Thus Q_g records the infinitesimal motion of the contact divisors.

Ampleness gives

$$H^1(\mathbb{P}^1, E_g) = H^1(\mathbb{P}^1, E_g(-2[0])) = 0.$$

Since Q_g is supported at infinity, $H^1(\mathbb{P}^1, Q_g) = 0$. Sequence (9.2) therefore gives $H^1(\mathbb{P}^1, g^*TX) = 0$ and makes the quotient map $H^0(\mathbb{P}^1, g^*TX) \to H^0(Q_g)$ surjective. Fixed-source deformation theory makes H smooth at g, with tangent space $H^0(\mathbb{P}^1, g^*TX)$. The quotient map is dc_g, so c is smooth at g, since B is smooth. Its fiber M is consequently smooth at g, with

$$T_gM = \ker dc_g = H^0(\mathbb{P}^1, E_g).$$

The supports of $2[0]$ and the contact divisors at infinity are disjoint, so $E_g|_{2[0]} = g^*TX|_{2[0]}$ under the natural inclusion. As in the proof of Lemma 8.2, dJ_g is the restriction map

$$H^0(\mathbb{P}^1, E_g) \longrightarrow H^0(2[0], E_g|_{2[0]}) \simeq T_{J(g)}TU.$$

It is surjective because $H^1(\mathbb{P}^1, E_g(-2[0])) = 0$. Since M and TU are smooth at the relevant points, J is smooth at g. □

Proof of Theorem 9.1. By Lemma 9.2, f belongs to the smoothness locus of the morphism $J : M \to TU$. Choose a nonempty Zariski-open neighborhood $S \subset M$ of f on which J is smooth. Then S is smooth. Let $\pi : TU \to U$ be the bundle projection. Evaluation at 0 is the smooth map $\mathrm{ev}_0 = \pi \circ J : S \to U$, and its image

$$U^+ = \mathrm{ev}_0(S)$$

is therefore nonempty and Zariski open.

For each $x \in U^+$ choose $g \in S$ over x. The analytification of (9.1) on $S \times \mathbb{C}$ and Lemma 8.1 yield centered entire families near x whose initial velocities form a frame. Hence U^+ lies in the LAP locus G_U of Definition 1.2. Since X is smooth and connected it is irreducible, and $U \setminus U^+$ is a proper closed algebraic, hence analytic, subset of U. Theorem 1.9 now shows that U is Oka. □

Remark 9.3. The positivity hypothesis in Theorem 9.1 has a necessary numerical consequence. Write

$$f^*T_X(-\log D) = \bigoplus_{\nu=1}^{n} \mathcal{O}_{\mathbb{P}^1}(a_\nu), \qquad a_\nu \geq 1.$$

The pushforward cycle $f_*[\mathbb{P}^1]$ counts the image curve with the degree of f onto its image. Since $\det T_X(-\log D) = \mathcal{O}_X(-(K_X + D))$, we obtain

$$-(K_X + D) \cdot f_*[\mathbb{P}^1] = \deg f^*T_X(-\log D) = \sum_{\nu=1}^{n} a_\nu \geq n.$$

The inequality constrains the logarithmic-curve criterion, not the Oka property of the complement; the critical-degree case in Subsection 9.2 makes this distinction explicit.

9.1.2. *Ample boundary and the log-Fano application.* An algebraic variety U is $\mathbb{A}^1$*-connected* if there is an algebraic family $a : T \times \mathbb{A}^1 \to U$ of nonconstant proper maps whose two-point evaluation

$$T \times \mathbb{A}^1 \times \mathbb{A}^1 \longrightarrow U \times U, \qquad (s, t_1, t_2) \longmapsto (a(s, t_1), a(s, t_2))$$

is dominant [14, Definitions 1.1–1.2].

Corollary 9.4. *Let X be a connected complex projective manifold with a reduced simple normal crossings divisor D. If $X \setminus D$ is $\mathbb{A}^1$-connected, then it is Oka.*

Proof. In characteristic zero the dominant two-point evaluation is separable. Chen–Zhu's very-free-curve criterion [14, Proposition 1.13(2)] gives a nonconstant map $f : \mathbb{P}^1 \to X$ with $f^{-1}(D) = \{\infty\}$ and $f^*T_X(-\log D)$ ample. Theorem 9.1 applies. □

Theorem 9.5. *Let X be a connected complex projective manifold of dimension at least two and $D \subset X$ a smooth irreducible ample divisor. If D is rationally connected, then $X \setminus D$ is Oka and holomorphically elliptic. In particular, the conclusion holds if $-(K_X + D)$ is ample.*

Proof. We use the following specialization of Chen–Zhu's fully-free-center criterion [14, Theorem 1.3 and Remark 1.4]: if a smooth complex projective variety of dimension at least two has a single smooth irreducible ample boundary divisor that is rationally connected, then its complement is $\mathbb{A}^1$-connected. To see how the hypotheses enter, the unique boundary stratum is D itself. Choose a very free curve $a : \mathbb{P}^1 \to D$. If $\dim D = 1$, use an isomorphism $\mathbb{P}^1 \simeq D$. Ampleness gives $\deg a^*\mathcal{O}_D(D) > 0$, so the contact intersection matrix consists of one positive

entry and has rank one. Thus D is fully free, as required by the criterion. Corollary 9.4 gives the Oka property.

Choose $k > 0$ with $\mathcal{O}_X(kD)$ very ample. In the associated projective embedding, the section defining kD cuts out a hyperplane. Consequently $X \setminus D$ is closed in the affine complement of that hyperplane. It is affine, hence Stein. A Stein Oka manifold is holomorphically elliptic [40], which proves the first assertion.

If $-(K_X + D)$ is ample, adjunction gives $-K_D = -(K_X + D)|_D$. Thus D is Fano and is rationally connected [11, 42]; the first assertion applies. □

Corollary 9.6. *Let $X \subset \mathbb{P}^N$ be a smooth complete intersection of multidegree $(d_1, \dots, d_c)$ and dimension at least two, and D be a smooth hypersurface section of degree $e \geq 1$. If*

$$d_1 + \cdots + d_c + e \leq N,$$

then $X \setminus D$ is Oka and holomorphically elliptic.

Proof. Let H denote the hyperplane class. Adjunction gives

$$-(K_X + D) = (N + 1 - d_1 - \cdots - d_c - e)H|_X,$$

which is ample under the stated inequality. The divisor D is also ample, since $D \sim eH|_X$.

Positive-dimensional projective complete intersections are connected. Hence X and D are connected, and smoothness makes D irreducible. Theorem 9.5 applies. □

9.2. **Smooth hypersurface complements.** We now determine exactly when the complement of a smooth degree-d hypersurface in $\mathbb{P}^n$ is Oka. The logarithmic construction already handles $d \leq n$ when $n \geq 2$. At the remaining positive degree $d = n+1$, the logarithmic tangent bundle has trivial determinant, so its pullback to a rational curve cannot be ample. Instead, we use lines with contact of order at least $d - 1$ with the hypersurface. Their complements are copies of $\mathbb{C}$ or $\mathbb{C}^*$, and their entire parametrizations vary holomorphically even when these two cases meet.

Kusakabe proved that complements of projective quadrics are Oka [44], and Xie established the Oka property and holomorphic ellipticity for smooth cubic complements in $\mathbb{P}^n$, $n \geq 2$ [56]. Theorem 1.11 gives the exact threshold in every degree. It answers, for smooth hypersurfaces, Forstnerič's question of which projective hypersurfaces have Oka complements [37].

The proof has three ingredients: generic projection makes successive polar equations transverse; these equations supply enough directions through a general point of the complement; and an exponential formula turns the directions into centered entire families. Analytic propagation then reaches the remaining points. Carlson–Griffiths' rank obstruction gives the converse [13].

9.2.1. *Successive polars and generic projection.* In the next two lemmas, N is the vector-space dimension and $X \subset \mathbb{P}^{N-1}$ denotes the hypersurface. In the application to $\mathbb{P}^n \setminus D$, these symbols mean $N = n + 1$ and $X = D$.

Successive polars encode high contact of lines with a hypersurface; see [5]. Use coordinates $v = (v_1, \dots, v_N)$ on $\mathbb{C}^N$. For a homogeneous polynomial F of degree d and a nonzero vector $a = (a_1, \dots, a_N)$, write

$$D_a = \sum_{i=1}^{N} a_i \frac{\partial}{\partial v_i}.$$

The equations $D_a^j F(v) = 0$ are bihomogeneous in (a, v), so their projective zero sets depend only on $[a]$ and $[v]$.

Suppose $d \geq 2$, $F(a) \neq 0$, and $v \neq 0$ satisfies $D_a^j F(v) = 0$ for $0 \leq j \leq d-2$. Write $dF_a(v)$ for the differential of F at a applied to v; by contrast, $D_a F(v)$ differentiates at v in the direction

a. For $s \in \mathbb{C}$, homogeneity gives

$$F(a+sv) = \sum_{\ell=0}^{d} \frac{s^{d-\ell}}{\ell!} D_a^\ell F(v) = F(a) + s\, dF_a(v). \tag{9.3}$$

The first equality follows for $s \neq 0$ by expanding $s^d F(v + a/s)$ and hence holds identically as a polynomial in s. The polar equations kill precisely the coefficients of $s^2, \dots, s^d$. Thus the restriction of F to the line through $[a]$ and $[v]$ has degree at most one in the affine parameter s, and the line meets the hypersurface with multiplicity at least $d-1$ at $[v]$.

For a general point $[a]$ of the complement, we therefore seek points $[v]$ satisfying these polar equations. Two properties are needed: transversality, so that the points $[v]$ can be continued holomorphically as $[a]$ moves, and spanning, so that the resulting line directions generate the tangent space at $[a]$.

Lemma 9.7. *Suppose $N \geq d \geq 3$ and $X = \{F = 0\} \subset \mathbb{P}^{N-1}$ is smooth. There is a nonempty Zariski-open subset $\Omega \subset \mathbb{P}^{N-1} \setminus X$ such that, for every $[a] \in \Omega$ and $1 \leq k \leq d-2$, the differentials of $D_a F, \dots, D_a^k F$ restricted to X are linearly independent at their common zeros. The differentials here are taken in local trivializations of the corresponding line bundles.*

Proof. The polar equations describe successive tangencies of projection from $[a]$. We first use generic projection to make those tangency conditions transverse, then identify their local equations with the polars. Fix a hyperplane $L \simeq \mathbb{P}^{N-2}$. Projection from $[a] \notin X \cup L$ gives a finite map $p_a : X \to L$: a positive-dimensional fiber would contain its whole line through $[a]$, contradicting $[a] \notin X$. Its differential has kernel of dimension at most one. The concrete generic-projection consequence we need is the following: for a general center, the critical locus is smooth of codimension one in X or empty; recursively, the locus where the kernel line is tangent to the preceding locus is smooth of codimension k in X or empty, for $2 \leq k \leq d-2$.

More precisely, we use the jet-transversality statement in Alzati–Ballico–Ottaviani [1, Section 2] for the smooth hypersurface X and point centers $T = \{[a]\}$: in their notation, $c = 1$ and $t = 0$. The center space is $\mathcal{F} = \mathbb{P}^{N-1} \setminus (X \cup L)$, and $\dim L = N-2 = \dim X$. For each k, their statement gives a nonempty Zariski-open subset $U_k \subset \mathcal{F}$ on which $j^k p_a$ is transverse to $\Sigma^{I_k} \subset J^k(X, L)$, where $I_k = (\underbrace{1, \dots, 1}_{k})$. This finite symbol prescribes the first k entries, with no terminal zero, so its inverse image includes points with further tangencies. For a corank-one map between manifolds of equal dimension, Σ^{I_k} has codimension k; the local equations below identify its inverse image with the successive tangency locus. The finite intersection $\Omega = \bigcap_{k=1}^{d-2} U_k$ is nonempty and Zariski open, since $\mathcal{F}$ is irreducible.

We identify the equations supplied by this transversality. Near a common zero $[v]$ of the polars, choose a linear form ℓ with $\ell(a) = 0$ and $\ell(v) \neq 0$. In the chart $\ell = 1$, choose coordinates (z, u, t) with $z \in \mathbb{C}^{N-3}$ so that translation by a is translation in t and the projection forgets t. If P is the equation of X in this chart, then

$$\partial_t^j P = (D_a^j F)|_{\ell=1}.$$

At the point under consideration $P_t = 0$. Smoothness lets us choose u with $P_u \neq 0$ and write X locally as $u = g(z, t)$. Set

$$A = P_u(z, g(z,t), t), \qquad q_j = (\partial_t^j P)(z, g(z,t), t), \qquad h_j = \partial_t^j g.$$

After shrinking, A is a unit. Differentiating $P(z, g(z,t), t) = 0$ repeatedly gives

$$q_j + A h_j \in (h_1, \dots, h_{j-1}), \qquad (q_1, \dots, q_j) = (h_1, \dots, h_j). \tag{9.4}$$

Indeed, the first identity starts with $q_1 + A h_1 = 0$, and $\partial_t q_j = q_{j+1} + P_{ut^j}(z, g(z,t), t) h_1$ proves the induction. The equality of ideals follows because A is invertible.

In these coordinates $p_a(z,t) = (z, g(z,t))$. Its critical locus is $S_1 = \{h_1 = 0\}$, and transversality gives $dh_1 \neq 0$ there. On this locus its kernel is spanned by ∂_t. Inductively, if $S_j = \{h_1 = \cdots = h_j = 0\}$ is the jth projection locus, then

$$\partial_t \in T_p S_j \quad \Longleftrightarrow \quad h_{j+1}(p) = 0, \qquad p \in S_j,$$

because $\partial_t h_i = h_{i+1}$. Thus the next corank-one locus is S_{j+1}. This identification also holds for the defining ideals: the Jacobian of $(p_a, h_1, \dots, h_j)$ has an identity block in the z variables, so its maximal-minor ideal is $(h_1, \dots, h_{j+1})$. Modulo the preceding equations the new equation is h_{j+1}. Thus the successive Thom–Boardman equations pull back to these generators, and jet transversality gives their differential rank $j+1$. Induction proves this rank statement up to $j = d-2$; when $d = 3$ only the first step is needed. At a common zero, (9.4) relates the differentials of the q_j and h_j by an invertible triangular matrix, so the same rank statement holds for the polars. □

Lemma 9.8. *Under the hypotheses and with the open set Ω of Lemma 9.7, set*

$$Z_a = \{F = D_a F = \cdots = D_a^{d-2} F = 0\} \subset \mathbb{P}^{N-1}.$$

Then Z_a is a nonempty smooth complete intersection of multidegree $(d, d-1, \dots, 2)$, dimension $N-d$, and degree $d!$. Its points span $\mathbb{P}^{N-1}$. Moreover, the incidence variety

$$\mathcal{I} = \{([a],[v]) \in \Omega \times X : [v] \in Z_a\}$$

is smooth, and its projection $\pi : \mathcal{I} \to \Omega$ is a surjective holomorphic submersion with fibers Z_a.

Proof. Let $S = \mathbb{C}[v_1, \dots, v_N]$ and $J = (F, D_a F, \dots, D_a^{d-2} F)$. There are $d-1 < N$ homogeneous generators. If the projective zero set were empty, the radical of J would be the irrelevant maximal ideal of height N, contrary to the height theorem. Thus Z_a is nonempty. The preceding lemma shows that the equations, including F, have independent differentials in projective coordinates along Z_a. Hence it is smooth of dimension $N-d$, and J has height $d-1$. The generators therefore form a regular sequence in the polynomial ring. This proves the complete-intersection assertion and gives

$$\deg Z_a = d(d-1)\cdots 2 = d!.$$

To prove spanning, let $\mathcal{I}_{Z_a}$ be the ideal sheaf of Z_a. Twist its Koszul resolution by $\mathcal{O}(1)$. Every line-bundle summand has negative degree, and the resolution has length $d-2 \leq N-2$, below the top cohomology degree of $\mathbb{P}^{N-1}$. The vanishings

$$H^i(\mathbb{P}^{N-1}, \mathcal{O}(-k)) = 0 \qquad (k > 0,\ 0 \leq i \leq N-2)$$

therefore give $H^0(\mathbb{P}^{N-1}, \mathcal{I}_{Z_a}(1)) = 0$. The complete-intersection scheme is smooth, hence reduced. A linear form vanishing at all its points belongs to this space and must be zero. Thus Z_a spans $\mathbb{P}^{N-1}$, including when $d = N$ and Z_a is zero-dimensional.

Finally, in local projective coordinates on $\Omega \times X$, the incidence equations have independent differentials in the X variables. The implicit function theorem makes $\mathcal{I}$ smooth and π a holomorphic submersion. Every fiber is nonempty, as already proved, so π is surjective. □

9.2.2. *Entire parametrizations of the line complements.* Buzzard–Lu [7, proof of Proposition 5.1, formulas (5.1)–(5.2)] use the following entire function in their parametrization of tangent-line complements of a smooth plane cubic. The entire function $E : \mathbb{C}^2 \to \mathbb{C}$ is given, for $(\lambda, t) \in \mathbb{C}^2$, by

$$E(\lambda, t) = \sum_{k=0}^{\infty} \frac{\lambda^k t^{k+1}}{(k+1)!} = \begin{cases} (e^{\lambda t} - 1)/\lambda, & \lambda \neq 0, \\ t, & \lambda = 0. \end{cases}$$

The series converges locally uniformly on $\mathbb{C}^2$ and gives

$$1 + \lambda E(\lambda, t) = e^{\lambda t}, \qquad \partial_t E(\lambda, 0) = 1.$$

For $\lambda \neq 0$ its image is $\mathbb{C} \setminus \{-1/\lambda\}$; for $\lambda = 0$ its image is all of $\mathbb{C}$.

Lemma 9.9. *Let F be a homogeneous polynomial on $\mathbb{C}^{n+1}$, $Y = \mathbb{P}^n \setminus \{F = 0\}$, and $V \subset Y$ an open set. Suppose there are nonzero holomorphic lifts $a, v_1, \dots, v_n : V \to \mathbb{C}^{n+1}$ such that $[a(y)] = y$, the classes of $v_1(y), \dots, v_n(y)$ form a basis modulo $\mathbb{C}a(y)$, and*

$$F(a(y) + sv_j(y)) = F(a(y)) + s\, dF_{a(y)}(v_j(y)) \quad (s \in \mathbb{C}). \tag{9.5}$$

Then every point of V admits centered entire families with initial velocities spanning its tangent space.

Proof. Set $c(y) = F(a(y))$ and $\lambda_j(y) = dF_{a(y)}(v_j(y))$. The required families are

$$h_j(y,t) = [a(y) + c(y)E(\lambda_j(y), t)v_j(y)].$$

The expression inside brackets has F-value $c(y)e^{\lambda_j(y)t} \neq 0$, so it is nonzero and defines a holomorphic map $h_j : V \times \mathbb{C} \to Y$. It is centered at y. For projectivization $q : \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{P}^n$,

$$\partial_t h_j(y, 0) = dq_{a(y)}(c(y)v_j(y)).$$

Since $\ker dq_{a(y)} = \mathbb{C}a(y)$ and $c(y) \neq 0$, these velocities form a basis. Lemma 2.1 now applies. □

For the polar directions this construction has a direct geometric meaning. Let $D = \{F = 0\}$. If $F(a) \neq 0$ and $[v] \in Z_a$, write $c = F(a)$ and $\lambda = dF_a(v)$. With homogeneous line coordinates $[r : s] \in \mathbb{P}^1$, homogenizing (9.3) gives

$$F(ra + sv) = r^{d-1}(rc + s\lambda).$$

The line $L_{a,v} = \{[ra+sv] : [r : s] \in \mathbb{P}^1\}$ is not contained in the hypersurface, and its intersection divisor is

$$D|_{L_{a,v}} = \begin{cases} (d-1)[v] + [\lambda a - cv], & \lambda \neq 0, \\ d[v], & \lambda = 0. \end{cases}$$

Consequently its complement is $\mathbb{C}^*$ or $\mathbb{C}$, respectively. The map $t \mapsto [a+cE(\lambda, t)v]$ parametrizes the entire line complement in either case and depends holomorphically on λ through zero.

In the line coordinates $[r : s]$, Figure 6 shows how the two intersection points coalesce.

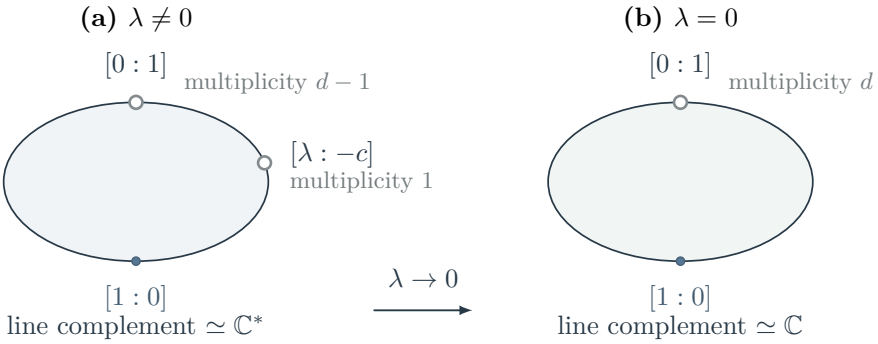


FIGURE 6. Intersection of a polar line with the hypersurface, in homogeneous line coordinates. Each oval represents $\mathbb{P}^1$ schematically; open marks are the deleted points, with their intersection multiplicities. Since $c \neq 0$, the simple point $[\lambda : -c]$ is distinct from $[0 : 1]$ for $\lambda \neq 0$ and coalesces with it at $\lambda = 0$. The base point $[1 : 0]$ remains in the complement. The same formula $t \mapsto [1 : cE(\lambda, t)]$ parametrizes both complements and stays holomorphic through the coalescence.

9.2.3. *The degree threshold.*

Proposition 9.10. *If $D \subset \mathbb{P}^n$ is a smooth hypersurface of degree $1 \le d \le n+1$, where $n \ge 1$, then $\mathbb{P}^n \setminus D$ is Oka.*

Proof. Write $D = \{F = 0\}$ and $Y = \mathbb{P}^n \setminus D$. The manifold Y is connected: a line joining two of its points is not contained in D and loses only finitely many points on removing D.

For $d = 1$, $Y \simeq \mathbb{C}^n$. For $d = 2$, the smooth quadric spans $\mathbb{P}^n$. Indeed, its quadratic form is nondegenerate and its isotropic vectors span $\mathbb{C}^{n+1}$: after diagonalization, the vectors $e_i \pm \mathrm{i}e_j$ are isotropic and span. This also covers $n = 1$, when the quadric is two distinct points. At any $y_0 = [a_0] \in Y$, choose constant isotropic vectors $v_1, \dots, v_n$ whose classes form a basis modulo $\mathbb{C}a_0$, together with a local lift $a(y)$ near y_0, and shrink its domain V so that this basis property persists. The quadratic identity

$$F(a + sv_j) = F(a) + s\,dF_a(v_j) + s^2F(v_j) = F(a) + s\,dF_a(v_j)$$

allows Lemma 9.9 to be applied at every point. The LAP criterion, Theorem 1.3, makes Y Oka.

Now suppose $3 \le d \le n+1$ and apply Lemma 9.8 with $N = n+1$ and $X = D$. For any $y_0 = [a_0] \in \Omega$, spanning by Z_{a_0} gives points $[v_1^0], \dots, [v_n^0] \in Z_{a_0}$ whose representatives form a basis modulo $\mathbb{C}a_0$. The submersion $\pi : \mathcal{I} \to \Omega$ has local holomorphic sections through $([a_0], [v_j^0])$. On a common neighborhood V of y_0, choose nonzero holomorphic lifts $a, v_1, \dots, v_n$ of these sections and of the inclusion $V \hookrightarrow \mathbb{P}^n$. Then

$$[a(y)] = y, \qquad [v_j(y)] \in Z_{a(y)},$$

and the basis property persists after shrinking V once. Identity (9.3) supplies (9.5). Lemma 9.9 therefore gives centered entire families at every point of Ω.

Let G_Y be the LAP locus of Definition 1.2. We have shown $\Omega \subset G_Y$. Since $Y \setminus \Omega$ is a proper closed algebraic subset of Y, Theorem 1.9 shows that Y is Oka throughout $d \le n+1$. □

Proof of Theorem 1.11. Proposition 9.10 proves the positive direction. Write $Y = \mathbb{P}^n \setminus D$. For $d \ge n+2$, the line bundle

$$\mathcal{O}_{\mathbb{P}^n}(K_{\mathbb{P}^n} + D) = \mathcal{O}_{\mathbb{P}^n}(d - n - 1)$$

is ample. Carlson–Griffiths' degeneracy theorem [13, Corollary 4.17] for the complement of a simple normal crossings divisor with positive log canonical class applies to the smooth divisor D. It says that every holomorphic map $\mathbb{C}^n \to Y$ has rank less than n everywhere. An Oka manifold is strongly dominable by Theorem 1.4, so Y cannot be Oka. This also treats $n = 1$ and $d \ge 3$.

Finally, D is ample, so Y is affine and hence Stein, by the affine-complement argument in Subsection 9.1. A Stein Oka manifold is holomorphically elliptic, and a holomorphically elliptic manifold is Oka [40]. Thus the two analytic properties are equivalent here, completing all three implications. □

Example 9.11. Smoothness is essential to the degree threshold. In homogeneous coordinates $[X : Y : Z]$ on $\mathbb{P}^2$, for every $m \ge 3$ the reduced plane divisor

$$D_m = \{Z(YZ^{m-1} - X^m) = 0\} \subset \mathbb{P}^2$$

has degree $m+1 > 3$ and is singular at $[0 : 1 : 0]$. Its complement lies in the chart $Z \ne 0$. In the affine coordinates $x = X/Z$ and $y = Y/Z$, the polynomial automorphism $(x, y) \mapsto (x, y - x^m)$ identifies it with $\mathbb{C} \times \mathbb{C}^*$. It is therefore Oka and holomorphically elliptic. This gives singular hypersurfaces with Oka complements in arbitrarily high degree.

10. Conjectures and further directions

A compact Kähler manifold is *special* in Campana's sense if it admits no meromorphic fibration onto a positive-dimensional base of general type with its induced orbifold structure [9, Definition 2.1].

Conjecture 10.1. *Every connected smooth complex projective manifold that is special in Campana's sense is Oka-1.*

Conjecture 10.2. *Every connected smooth complex projective Oka-1 manifold is Oka.*

Conjecture 10.3. *Let $\pi : \mathcal{X} \to S$ be a smooth projective morphism of complex manifolds, with connected base S and connected fibers. If one fiber of π is Oka, then every fiber is Oka.*

The first two conjectures together would make every special smooth projective manifold Oka, answering this case of Campana–Winkelmann's question [10, Question 1.8]; see also Forstnerič [37, Problem 11.3]. Related Oka-1 and deformation questions appear in [2, Problem 2.11(c)] and [28, Problem F].

Every compact Kähler Oka manifold is special [9, Corollary 8.11]. Thus the first two conjectures would identify the special, Oka-1, and Oka classes in the projective category. Together with Campana's conjecture that specialness is invariant under smooth deformation [9, Conjecture 6.23], they would imply Conjecture 10.3.

Evidence includes rationally connected manifolds (Theorem 1.10), K3 surfaces [49, Theorem 1.1], and the Schoen fiber products with disjoint singular-value sets and sufficiently general Gross–Popescu Calabi–Yau threefolds proved Oka in [3, Theorems 8.7–8.8]. Analogous questions can be asked for compact Kähler manifolds.

Appendix A. The local extension condition and its equivalence with LAP

We use the local planar approximation condition of OpenAI [49, Definition 2.1] and prove that it is equivalent to LAP at each point.

Definition A.1. Let Y be a complex manifold of dimension $n \geq 1$. One says that Y satisfies $\mathcal{L}$ at $x \in Y$ if there is an open neighborhood $V \ni x$ such that, for every special planar pair $D \subset D'$, every integer $q \geq 0$, every pair of closed polydiscs $P_0 \Subset \operatorname{Int} P \subset \mathbb{C}^q$, and every holomorphic map f from a neighborhood of $D \times P$ into V, the restriction $f|_{D\times P_0}$ is uniformly approximable by maps holomorphic near $D' \times P_0$ with values in Y. The set V is called a witness neighborhood. The manifold Y satisfies $\mathcal{L}$ if it does so at every point.

Special planar pairs are as in Definition 3.1. Passive parameters are retained through the following notational reformulation. If f is holomorphic near $D \times Q$ into V, compactness permits a slightly larger closed polydisc Q^+ such that $Q \Subset \operatorname{Int} Q^+$ and f is still holomorphic near $D \times Q^+$ into V. Applying $\mathcal{L}$ with $P_0 = Q$ and $P = Q^+$ therefore retains the whole prescribed closed polydisc Q. Conversely, this retained-parameter formulation immediately implies the definition by restriction.

The next lemma is the disc-enlargement step of OpenAI [49, Proposition 2.9], in arbitrary target dimension.

Lemma A.2. *Let V be a witness neighborhood for $\mathcal{L}$ and f holomorphic near $\overline{\Delta}_r \times P$ into V, where $r > 0$ and P is a closed polydisc. For every $S > r$, the map f is uniformly approximable on $\overline{\Delta}_r \times P$ by maps holomorphic near $\overline{\Delta}_S \times P$ with values in Y.*

Proof. Use the construction in the proof of Lemma 5.2, with V for every boundary arc and the retained-parameter formulation of $\mathcal{L}$ in place of local planar extension. Compactness allows the old chart spray and slightly enlarged passive products to take values in V, so the sector collars also have their spray images in V. Including the original passive variables and the

spray variables in Q, the extension and simultaneous center gluing retain a neighborhood of the prescribed P and give the required approximation. □

Proposition A.3. *For every positive-dimensional complex manifold Y and every $x \in Y$, the conditions $\mathcal{L}$ and* LAP *at x are equivalent.*

Proof. Suppose first that LAP holds at x. Lemma 3.4 provides a neighborhood W_x on which local planar extension retains every closed passive polydisc. Restricting to P_0 gives $\mathcal{L}$ with witness W_x.

Conversely, let V witness $\mathcal{L}$ at x, and choose a coordinate parametrization ψ about $0 \in \mathbb{C}^n$, with $\psi(0) = x$ and image in V. For each $1 \le j \le n$, regard the jth coordinate as the scalar variable z and the remaining coordinates as passive variables p. After restriction, this gives biholomorphic parametrizations ψ_j near fixed products $\overline{\Delta}_r \times P_j$, where $0 \in \operatorname{Int} P_j$, and the vectors $\partial_z \psi_j(0,0)$ form a basis of $T_x Y$.

Choose $S_\nu \to \infty$. Lemma A.2 gives maps

$$\sigma_{j,\nu} : \operatorname{Op}(\overline{\Delta}_{S_\nu} \times P_j) \longrightarrow Y$$

whose errors relative to ψ_j on $\overline{\Delta}_r \times P_j$ tend to zero. The products P_j remain fixed as ν increases. Lemma 7.1 now supplies the exactly centered families required for LAP at x. □

Acknowledgments

The authors thank Franc Forstnerič for suggesting the name *local approximation property* (LAP) and for his valuable comments on an earlier version of this paper.

Grant support

S.-Y. Xie acknowledges partial support from the National Key R&D Program of China under Grants No. 2023YFA1010500 and No. 2021YFA1003100, and from the National Natural Science Foundation of China under Grants No. 12288201 and No. 12471081, as well as support from the Xiaomi Young Talents Program.

Statement on the use of artificial intelligence

This manuscript grew out of earlier drafts, working notes, and discussions within the research group. ChatGPT-6-Astra was used in exploring and organizing the arguments and in revising the manuscript. The authors reviewed the text and take full responsibility for its mathematical content and final form.

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(Sicheng An) School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, China

Email address: ansicheng23@mails.ucas.ac.cn

(Yun-Heng Du) Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China

Email address: duyunheng22@mails.ucas.ac.cn

(Bin Guo) Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China

Email address: guobin181@mails.ucas.ac.cn

(Peng-Chao Wang) Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China

Email address: wangpengchao26@mails.ucas.ac.cn

(Song-Yan Xie) State Key Laboratory of Mathematical Sciences, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China; School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, China

Email address: xiesongyan@amss.ac.cn